

# Unit-II: Statistics and Exploratory Data Analysis

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# 1 How to Use This Handout

This handout have the complete **Unit 2 (Statistics & EDA)** material. It is organised into six parts:

1. **Statistical Foundations** – the vocabulary, thinking, and probability required for data-driven engineering.
2. **Data Cleaning & Preparation** – the unglamorous 60-80% of every real project.
3. **Exploratory Data Analysis (EDA)** – the workflow, the Iris example, and the visual grammar of ggplot2.
4. **Correlation & Feature Relationships** – how to spot signal vs. redundancy.
5. **Regression Analysis** – simple, multiple, polynomial, logistic, LDA, ridge, lasso, quantile.
6. **Two Case Studies** – rainfall variability and retail sales, each end-to-end.

**How to read this handout:** Read a section, then run its code chunk in RStudio (or knit the file). Change the numbers. Break the data on purpose. Watch what the model does. That is how engineers learn machines – by poking them until they behave.

---

## 2 Part I – Statistical Foundations

### 2.1 What is Statistics?

**Statistics** is the science of collecting, organizing, summarizing, analyzing, and interpreting data to make informed decisions under uncertainty. In engineering, statistics is used for:

- **Quality control** – monitoring manufacturing defects.
- **Reliability analysis** – predicting failure times of components.
- **Experimental design** – testing which factors affect performance.
- **Risk assessment** – estimating probability of extreme events (floods, earthquakes).
- **Process optimization** – reducing waste and improving yield.

**Key definition:** Statistics is not just about numbers – it is a framework for **decision-making under uncertainty**. Engineers use it to separate signal from noise.

#### 2.1.1 Descriptive vs. Inferential Statistics

| Type               | Purpose                                           | Example                                    |
|--------------------|---------------------------------------------------|--------------------------------------------|
| <b>Descriptive</b> | Summarize and describe the data at hand           | Mean rainfall of Sydney = 85 mm            |
| <b>Inferential</b> | Draw conclusions about a population from a sample | Is Sydney's rainfall increasing over time? |

#### 2.1.2 Population vs. Sample

- **Population** – the complete set of all items of interest (e.g., all rainfall days in Sydney's history).
- **Sample** – a subset of the population actually measured (e.g., 800 days of rainfall data).
- **Parameter** – a numerical summary of a population (e.g., true mean rainfall  $\mu$ ).
- **Statistic** – a numerical summary of a sample (e.g., sample mean  $\bar{x}$ ).

**Exam tip:** A statistic *estimates* a parameter. The sample mean  $\bar{x}$  estimates the population mean  $\mu$ . The sample standard deviation  $s$  estimates the population standard deviation  $\sigma$ .

## 2.2 Statistical Learning

**Statistical learning** is the set of tools for modeling and understanding complex datasets. It blends statistics with machine learning.

Statistical learning = using data to estimate the unknown function  $f$  that maps inputs  $X$  to output  $Y$ .

We observe a quantitative response  $Y$  and  $p$  predictors  $X_1, \dots, X_p$ , and assume:

$$Y = f(X) + \varepsilon$$

where  $f$  is a fixed but unknown function and  $\varepsilon$  is random noise with mean zero.

### 2.2.1 Supervised vs. Unsupervised Learning

| Aspect                 | Supervised Learning             | Unsupervised Learning         |
|------------------------|---------------------------------|-------------------------------|
| <b>Inputs</b>          | $X$ and $Y$ (labeled data)      | Only $X$ (no labels)          |
| <b>Goal</b>            | Predict $Y$ from $X$            | Discover structure in $X$     |
| <b>Examples</b>        | Regression, classification, LDA | Clustering, PCA               |
| <b>Engineering use</b> | Predicting strength of concrete | Grouping similar soil samples |

### 2.2.2 Parametric vs. Non-Parametric Methods

**Parametric** methods assume a functional form for  $f$  (e.g., linear). They require fewer observations and are easy to interpret, but the assumed form may not match reality.

**Non-parametric** methods make no explicit assumptions. They are flexible but require many more observations and risk overfitting.

**Overfitting:** When a model follows the random noise in training data too closely, it performs poorly on new data. Flexible models with many parameters are prone to overfitting.

### 2.2.3 Prediction vs. Inference

- **Prediction:** use  $\hat{Y} = \hat{f}(X)$  when  $Y$  is unavailable.
- **Inference:** understand *how*  $Y$  changes with  $X_1, \dots, X_p$  – the *form* of  $f$  matters, not just prediction accuracy.

### 2.2.4 The Bias–Variance Trade-off

- High bias = model too simple = **underfitting**.
- High variance = model too complex = **overfitting**.
- The goal is the **sweet spot** – a model that captures signal without fitting noise.

---

## 2.3 R and Data Structures

**R** is a free, open-source language for statistical computing and graphics.

- **1992:** Ross Ihaka and Robert Gentleman, University of Auckland.
- **1997:** CRAN established.
- **Today:** 20,000+ packages on CRAN for statistics, ML, visualization, geospatial, and engineering.

### 2.3.1 Types of Engineering Data

| Data Type        | Description                    | Example                                   |
|------------------|--------------------------------|-------------------------------------------|
| <b>Numerical</b> | Continuous or discrete numbers | Temperature = 23.5°C,<br>Rainfall = 12 mm |

| Data Type          | Description                      | Example                          |
|--------------------|----------------------------------|----------------------------------|
| <b>Categorical</b> | Labels or groups                 | Soil type = {Clay, Sand, Silt}   |
| <b>Time-series</b> | Observations indexed by time     | Daily rainfall 2011–2014         |
| <b>Spatial</b>     | Data with geographic coordinates | Rainfall at weather stations     |
| <b>Ordinal</b>     | Ordered categories               | Concrete grade = {M20, M25, M30} |
| <b>Binary</b>      | Two categories                   | Yes/No, Pass/Fail                |

### 2.3.2 R Data Structures

|                   | Structure | Description                    | Mixed types? |
|-------------------|-----------|--------------------------------|--------------|
| <b>Vector</b>     |           | 1D sequence of same type       | No           |
| <b>Factor</b>     |           | Categorical variable           | No (levels)  |
| <b>Matrix</b>     |           | 2D, same type                  | No           |
| <b>Data frame</b> |           | 2D, columns can differ         | Yes          |
| <b>List</b>       |           | Ordered collection of anything | Yes          |
| <b>Tibble</b>     |           | Modern data frame              | Yes          |

A **data frame** is the workhorse of R data analysis – it is like an Excel sheet where each column can be a different type (numeric, character, factor, date).

## 2.4 Descriptive Statistics

### 2.4.1 Central Tendency

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (\text{mean})$$

- **Median** – middle value when sorted. Robust to outliers.
- **Mode** – most frequent value.

When data is skewed (rainfall, income), the **median** is a better measure of centre. When symmetric, mean = median = mode.

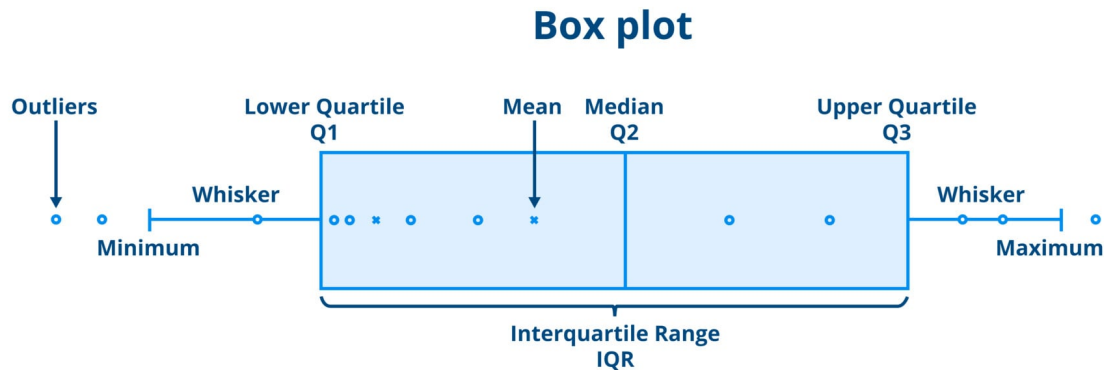
### 2.4.2 Measures of Spread

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad s = \sqrt{s^2}$$

$$\text{IQR} = Q_3 - Q_1 \quad \text{Range} = \text{Max} - \text{Min}$$

### 2.4.3 Five-Number Summary

1. Minimum
2. Q1 (25th percentile)
3. Median (Q2)
4. Q3 (75th percentile)
5. Maximum



This is the basis of the **boxplot**.

### 2.4.4 Shape of Distribution

| Shape               | Description     | Mean vs. Median |
|---------------------|-----------------|-----------------|
| <b>Symmetric</b>    | Bell-shaped     | Mean = Median   |
| <b>Right-skewed</b> | Long right tail | Mean > Median   |
| <b>Left-skewed</b>  | Long left tail  | Mean < Median   |

**Rainfall example:** Most days have little rain, but a few days have downpours → **right-skewed**. The mean is pulled upward by extremes, so the median is a better summary.

### 2.4.5 Coefficient of Variation, Standard Error, Confidence Intervals

$$CV = \frac{s}{\bar{x}} \times 100\% \quad SE = \frac{s}{\sqrt{n}} \quad \bar{x} \pm 1.96 \times SE \text{ (95\% CI)}$$

## 2.5 Probability Foundations for Risk Assessment

### 2.5.1 Basic Rules

- **Complement:**  $P(A^c) = 1 - P(A)$
- **Addition:**  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- **Multiplication (independent):**  $P(A \cap B) = P(A) \times P(B)$
- **Conditional:**  $P(A|B) = P(A \cap B)/P(B)$

### 2.5.2 Common Distributions

|                    | Distribution | Type | Use case                            |
|--------------------|--------------|------|-------------------------------------|
| <b>Binomial</b>    | Discrete     |      | Number of successes in $n$ trials   |
| <b>Poisson</b>     | Discrete     |      | Number of events in a time interval |
| <b>Normal</b>      | Continuous   |      | Heights, errors, temperatures       |
| <b>Exponential</b> | Continuous   |      | Time between events                 |
| <b>Log-normal</b>  | Continuous   |      | Rainfall, income                    |
| <b>Weibull</b>     | Continuous   |      | Reliability, failure times          |

**Return periods:** A 100-year flood has a **1%** chance of occurring in any given year. This is based on probability distributions fitted to historical data.

---

## 3 Part II – Data Cleaning & Preparation

**Civil connection:** You never build on an uncleared site. You remove debris, level the ground, and set out the grid before the first column is cast. Data preparation is exactly this – and in practice it takes **60–80%** of a data scientist's time.

### 3.1 Handling Missing Values

Missing values can skew analysis and cause errors in statistical modeling. Common strategies: **remove rows** or **impute** with mean/median/mode.

#### 3.1.1 Demonstrating on the Iris Dataset

```
# Artificially introduce NA values for teaching
iris_missing <- iris
iris_missing$Sepal.Length[c(5, 15, 30, 50, 70, 130)] <- NA
iris_missing$Petal.Width[c(10, 40, 53, 81, 132)] <- NA

# Count missing values per column
colSums(is.na(iris_missing))
```

```
## Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##           6           0           0           5           0
```

#### 3.1.2 Visualising the Missingness Pattern

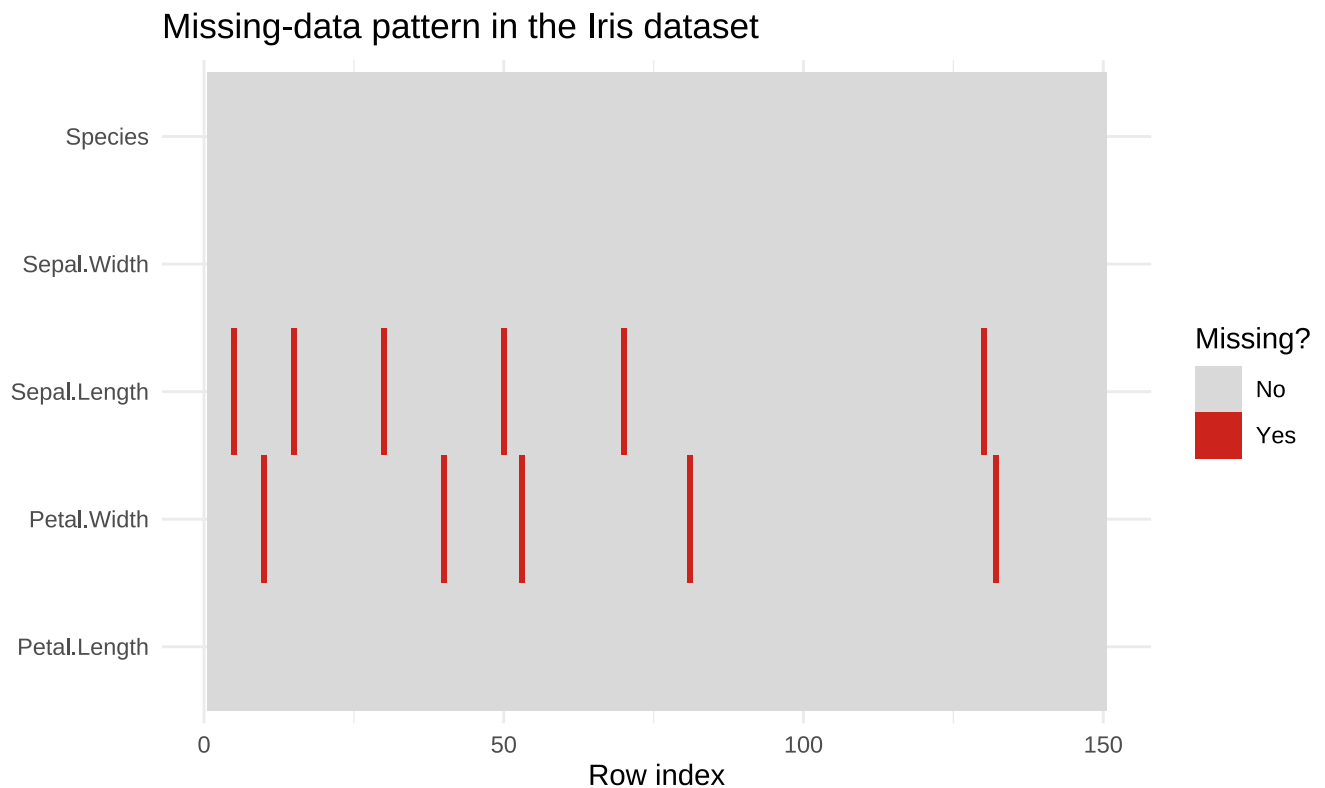
```
missing_matrix <- as.data.frame(is.na(iris_missing))
missing_matrix$row_num <- seq_len(nrow(missing_matrix))
missing_long <- tidyr::pivot_longer(
  missing_matrix, ~row_num,
```

```

names_to = "Variable", values_to = "Is_Missing"
)

ggplot(missing_long, aes(x = row_num, y = Variable, fill = Is_Missing)) +
  geom_raster() +
  scale_fill_manual(
    values = c("TRUE" = "#cc241d", "FALSE" = "grey85"),
    name = "Missing?", labels = c("No", "Yes")
  ) +
  labs(
    title = "Missing-data pattern in the Iris dataset",
    x = "Row index", y = NULL
  ) +
  theme_minimal(base_size = 11)

```



**How to read this figure:** Each red square is a missing value. Red cells are scattered randomly – consistent with *Missing Completely At Random* (MCAR). If they clustered (e.g., only in one species), we would suspect a systematic cause.

### 3.1.3 Imputation Strategy

```

iris_imputed <- iris_missing %>%
  mutate(
    Sepal.Length = ifelse(is.na(Sepal.Length),

```

```

    median(Sepal.Length, na.rm = TRUE),
    Sepal.Length
  ),
  Petal.Width = ifelse(is.na(Petal.Width),
    median(Petal.Width, na.rm = TRUE),
    Petal.Width
  )
)
colSums(is.na(iris_imputed))

```

```

## Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##           0           0           0           0           0

```

### Drop vs. impute?

- **Drop** when only a few rows are missing (say < 5%) and you have plenty of data.
- **Impute** when data is precious – expensive field tests, rare flood events, long-term monitoring.
- Use the **median** rather than the mean for skewed columns, because one wild value can drag the mean far away while the median barely moves.

---

## 3.2 Removing Duplicates

Duplicate rows over-represent certain observations and bias results.

```

set.seed(1)
iris_dup <- rbind(iris, iris[sample(150, 8), ]) # add 8 duplicates
cat("Rows with duplicates:", nrow(iris_dup), "\n")

```

```
## Rows with duplicates: 158
```

```

iris_unique <- dplyr::distinct(iris_dup)
cat("Rows after removing duplicates:", nrow(iris_unique), "\n")

```

```
## Rows after removing duplicates: 149
```

**Caution:** Not all duplicates are errors. Multiple orders from the same customer on the same day may be legitimate. Understand the context before removing.

---

### 3.3 Outlier Detection

#### 3.3.1 IQR Method

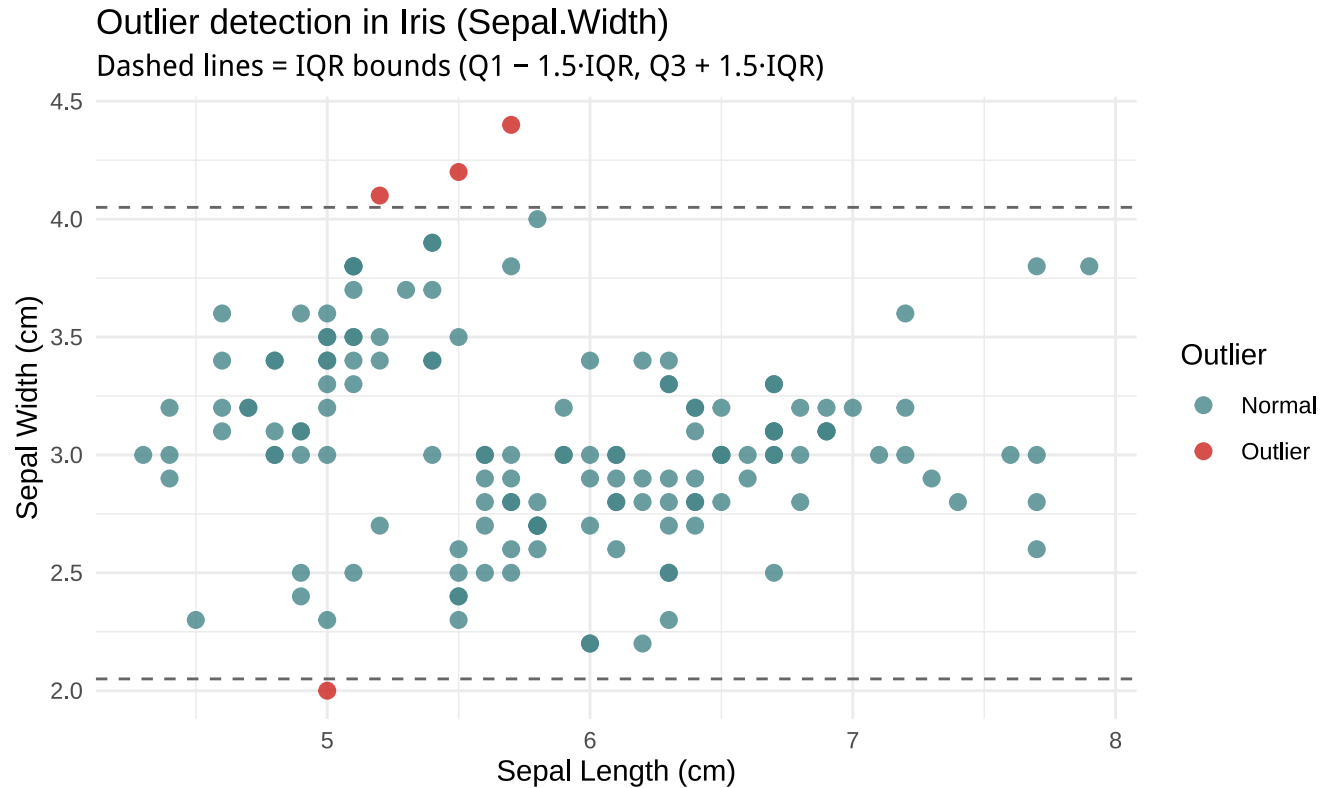
An observation is an outlier if:

$$x < Q_1 - 1.5 \times \text{IQR} \quad \text{or} \quad x > Q_3 + 1.5 \times \text{IQR}$$

```
Q1 <- quantile(iris$Sepal.Width, 0.25)
Q3 <- quantile(iris$Sepal.Width, 0.75)
IQR_val <- Q3 - Q1
lower <- Q1 - 1.5 * IQR_val
upper <- Q3 + 1.5 * IQR_val

iris_out <- iris %>%
  mutate(Outlier = ifelse(Sepal.Width < lower | Sepal.Width > upper,
    "Outlier", "Normal"
  ))

ggplot(iris_out, aes(Sepal.Length, Sepal.Width, color = Outlier)) +
  geom_point(size = 2.5, alpha = 0.8) +
  scale_color_manual(values = c("Normal" = "#458588", "Outlier" = "#cc241d")) +
  geom_hline(
    yintercept = c(lower, upper), linetype = "dashed",
    color = "grey40"
  ) +
  labs(
    title = "Outlier detection in Iris (Sepal.Width)",
    subtitle = "Dashed lines = IQR bounds (Q1 - 1.5·IQR, Q3 + 1.5·IQR)",
    x = "Sepal Length (cm)", y = "Sepal Width (cm)"
  ) +
  theme_minimal(base_size = 11)
```



### 3.3.2 Z-Score Method

$$z = \frac{x - \bar{x}}{s}$$

- Observation is an outlier if  $|z| > 3$  (or sometimes 2.5).
- Assumes normality; sensitive to the very outliers it tries to detect.

**In rainfall data, extreme values (storms) are often real and important.** Removing them would destroy the analysis of extreme events. Always investigate before removing outliers.

## 4 Part III – Exploratory Data Analysis (EDA)

### 4.1 What is EDA?

**Exploratory Data Analysis** is the process of examining data before formal modeling to:

- Understand distributions.
- Detect patterns, anomalies, and relationships.
- Form hypotheses.
- Guide model selection.

EDA is detective work. You look for clues before building a case (model).

#### 4.1.1 The EDA Workflow

1. **Import data** – CSV, Excel, or database.
2. **Inspect structure** – dimensions, types, head, tail.
3. **Check missing values** – count and visualize.
4. **Clean data** – missing, duplicates, outliers.
5. **Univariate analysis** – one variable at a time.
6. **Bivariate analysis** – relationships between pairs.
7. **Multivariate analysis** – correlations, interactions.
8. **Visualise** – plots for each finding.
9. **Interpret** – conclusions and hypotheses.

#### 4.1.2 Univariate Analysis

- **Numerical:** histogram, density plot, boxplot, summary statistics.
- **Categorical:** bar chart, frequency table.

#### 4.1.3 Bivariate Analysis

- **Num vs. Num:** scatter plot, correlation.
- **Num vs. Cat:** boxplot, violin plot, grouped summary.
- **Cat vs. Cat:** contingency table, stacked bar chart.

#### 4.1.4 Multivariate Analysis

- Scatterplot matrix (pairs plot).
- Correlation heatmap.
- Faceting (small multiples).
- Colouring by group.

---

## 4.2 The Iris Dataset – A Classic Teaching Tool

The **Iris dataset** (Fisher, 1936) contains 150 flowers from three species:

- *Iris setosa*, *Iris versicolor*, *Iris virginica* (50 each).

Four measurements per flower (in cm): Sepal.Length, Sepal.Width, Petal.Length, Petal.Width.

#### 4.2.1 Botanical Background

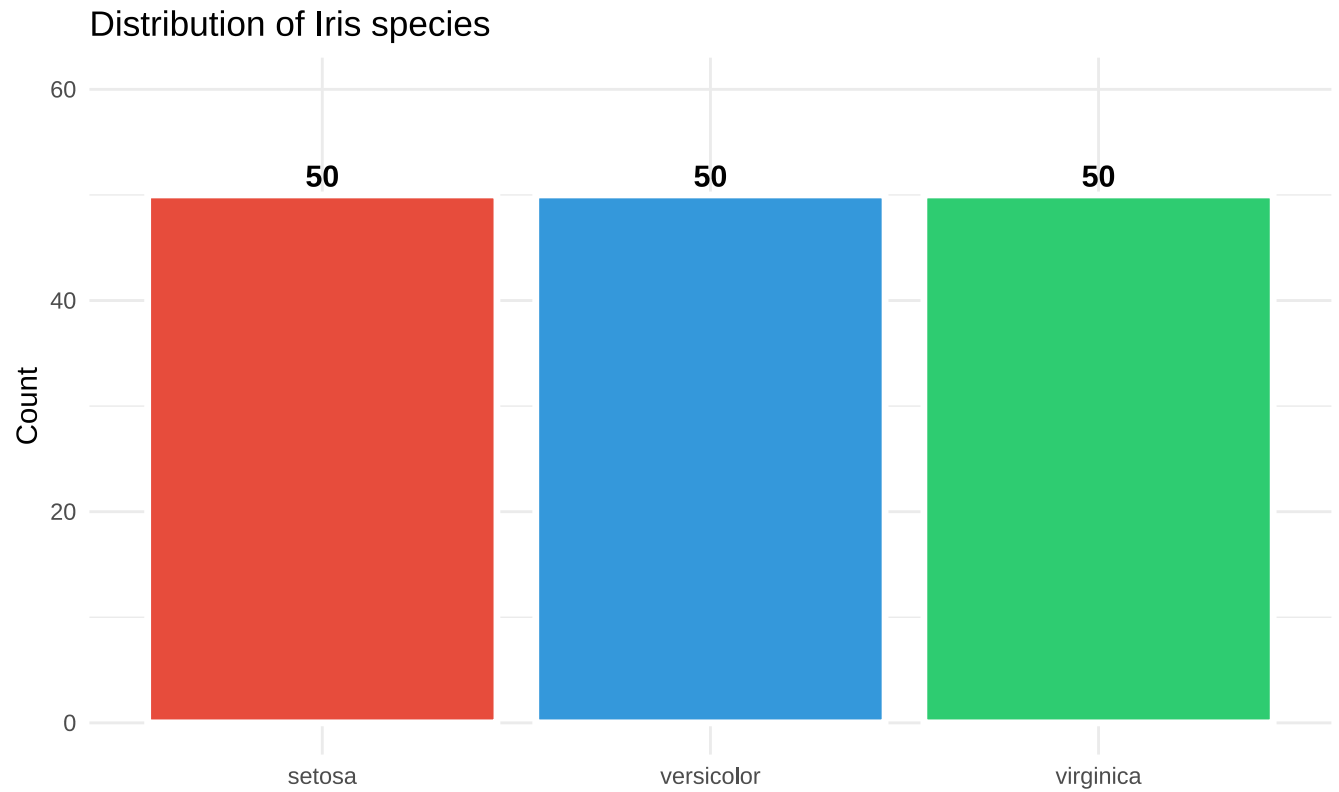
| Term         | Description                            | Function                        |
|--------------|----------------------------------------|---------------------------------|
| <b>Sepal</b> | Outer, leaf-like protective structures | Protect flower bud before bloom |
| <b>Petal</b> | Inner, often colourful structures      | Attract pollinators             |

#### 4.2.2 First Look

```
summary(iris)
```

```
##   Sepal.Length   Sepal.Width   Petal.Length   Petal.Width
##   Min.    :4.300   Min.    :2.000   Min.    :1.000   Min.    :0.100
##   1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300
##   Median :5.800   Median :3.000   Median :4.350   Median :1.300
##   Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
##   3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
##   Max.    :7.900   Max.    :4.400   Max.    :6.900   Max.    :2.500
##           Species
##   setosa      :50
##   versicolor :50
##   virginica   :50
##
##
##
```

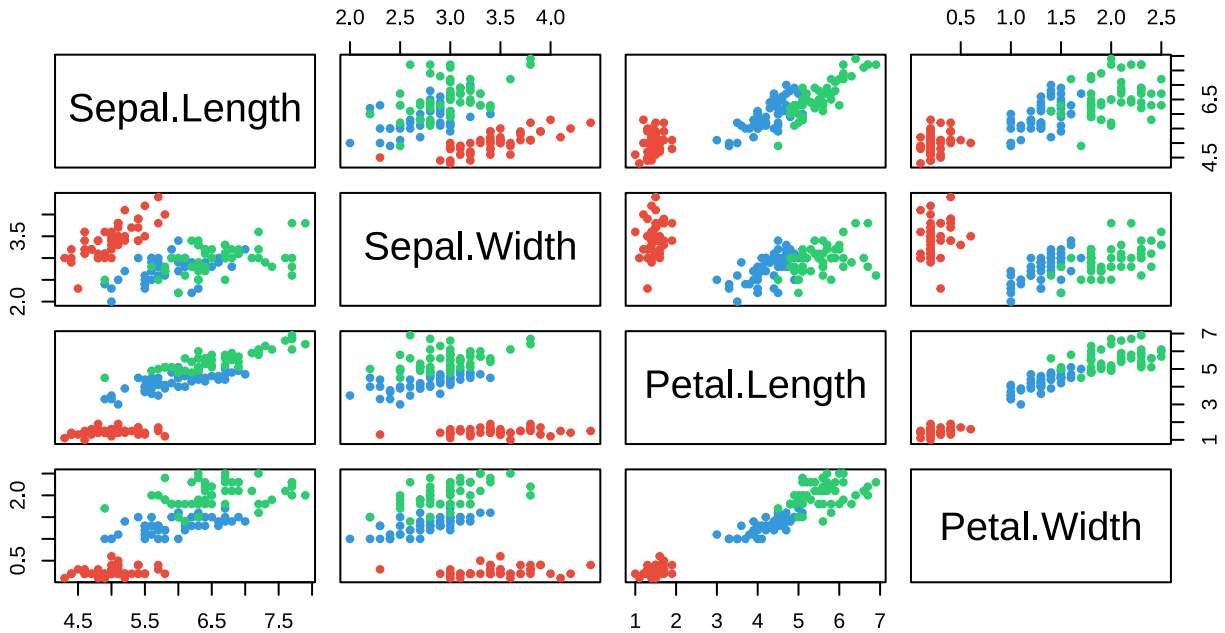
```
ggplot(iris, aes(Species, fill = Species)) +
  geom_bar(color = "white", linewidth = 1.2) +
  geom_text(
    stat = "count", aes(label = after_stat(count)),
    vjust = -0.4, size = 4, fontface = "bold"
  ) +
  scale_fill_manual(values = c("#E74C3C", "#3498DB", "#2ECC71")) +
  labs(title = "Distribution of Iris species", x = NULL, y = "Count") +
  theme_minimal(base_size = 11) +
  theme(legend.position = "none") +
  ylim(0, 60)
```



#### 4.2.3 Scatterplot Matrix

```
pairs(iris[, 1:4],  
      col = c("#E74C3C", "#3498DB", "#2ECC71")[iris$Species],  
      pch = 19, cex = 0.7,  
      main = "Scatterplot matrix of Iris measurements"  
)
```

## Scatterplot matrix of Iris measurements



**How to read the pairs plot:** Each off-diagonal cell is a scatter of two measurements. Notice **Petal.Length** vs. **Petal.Width** cleanly separates *setosa* (red) from the other two. Sepals are less discriminating – a key insight for feature selection.

### 4.3 Visual Analytics with ggplot2

**ggplot2** implements the *Grammar of Graphics* – a systematic, layer-by-layer approach. Every plot has:

1. **Data** – the dataset.
2. **Aesthetics (aes)** – map variables to visual properties.
3. **Geoms** – geometric objects (`geom_point()`, `geom_bar()`, ...).
4. **Scales** – control mapping from data to visual space.
5. **Facets** – sub-plots by categorical variables.
6. **Themes** – non-data elements (fonts, colours, gridlines).

#### 4.3.1 Scatter Plots (`geom_point`)

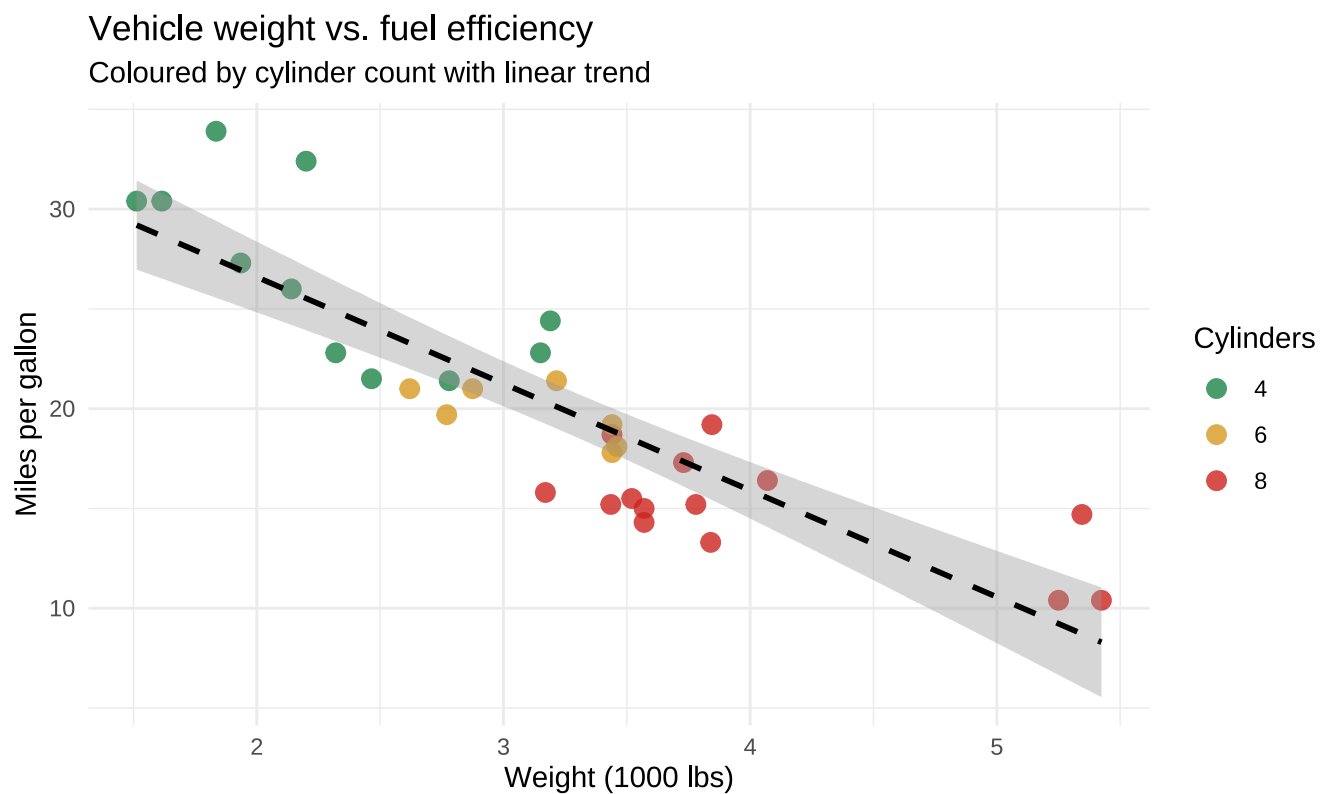
**Use for:** relationship between two continuous variables.

```
ggplot(mtcars, aes(wt, mpg)) +  
  geom_point(aes(color = factor(cyl)), size = 3, alpha = 0.8) +  
  geom_smooth(  
    method = "lm", se = TRUE, color = "black",
```

```

linetype = "dashed"
) +
scale_color_manual(
  values = c("#1e8449", "#d79921", "#cc241d"),
  name = "Cylinders"
) +
labs(
  title = "Vehicle weight vs. fuel efficiency",
  subtitle = "Coloured by cylinder count with linear trend",
  x = "Weight (1000 lbs)", y = "Miles per gallon"
) +
theme_minimal(base_size = 11)

```



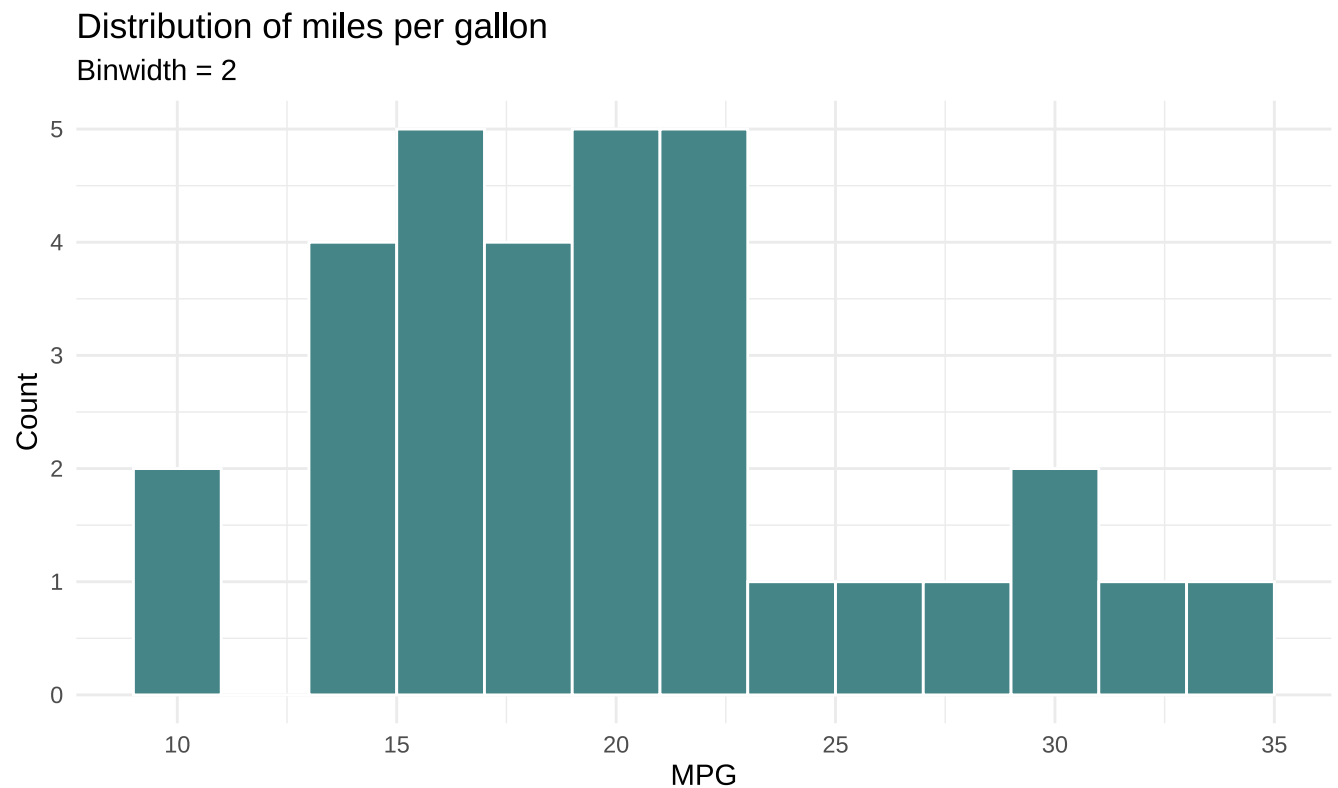
#### 4.3.2 Histograms (geom\_histogram)

Use for: distribution of a single continuous variable.

```

ggplot(mtcars, aes(mpg)) +
  geom_histogram(binwidth = 2, fill = "#458588", color = "white") +
  labs(
    title = "Distribution of miles per gallon",
    subtitle = "Binwidth = 2",
    x = "MPG", y = "Count"
  ) +
  theme_minimal(base_size = 11)

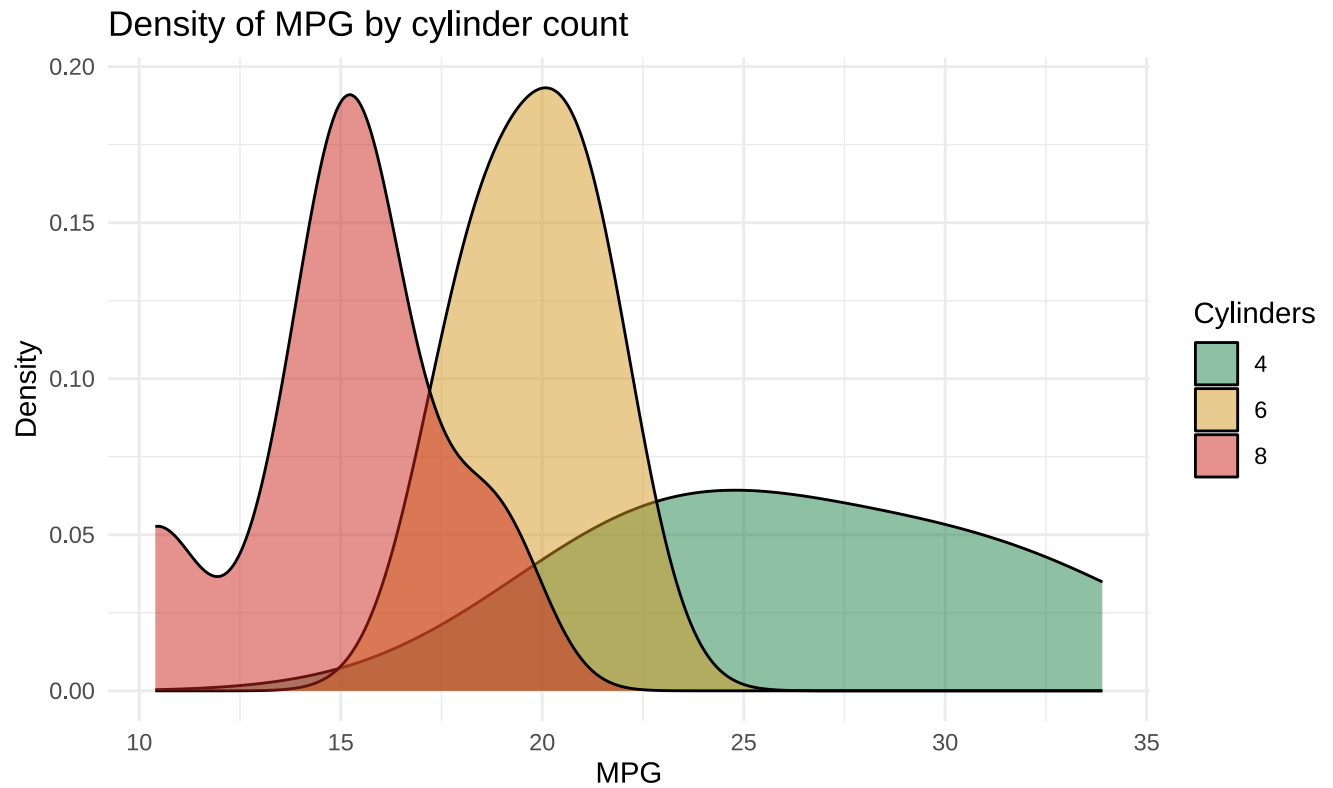
```



#### 4.3.3 Density Plots (`geom_density`)

**Use for:** smoothed distribution, especially good for comparing groups.

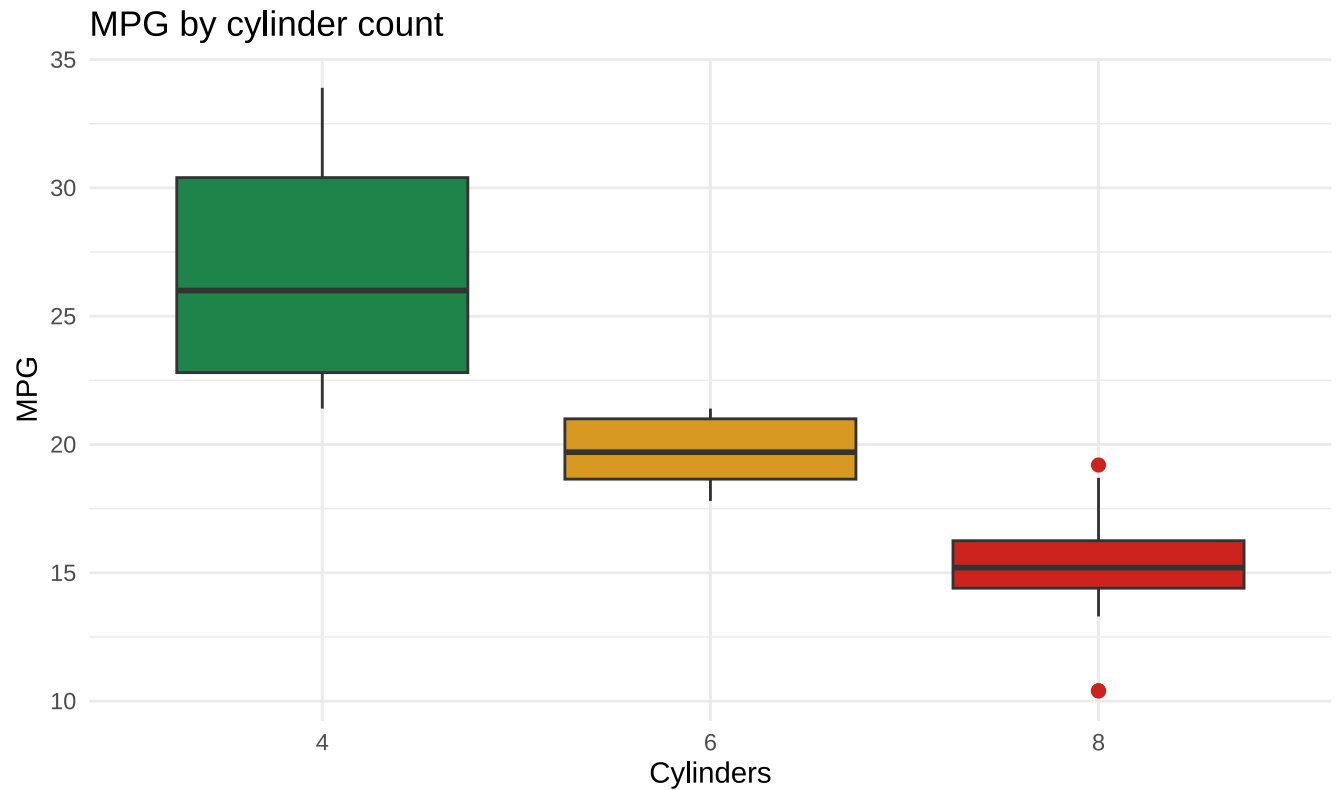
```
ggplot(mtcars, aes(mpg, fill = factor(cyl))) +  
  geom_density(alpha = 0.5, adjust = 1.5) +  
  scale_fill_manual(  
    values = c("#1e8449", "#d79921", "#cc241d"),  
    name = "Cylinders"  
  ) +  
  labs(  
    title = "Density of MPG by cylinder count",  
    x = "MPG", y = "Density"  
  ) +  
  theme_minimal(base_size = 11)
```



#### 4.3.4 Boxplots (geom\_boxplot)

**Use for:** five-number summary + outliers; comparing distributions across groups.

```
ggplot(mtcars, aes(factor(cyl), mpg, fill = factor(cyl))) +  
  geom_boxplot(outlier.color = "#cc241d", outlier.size = 2) +  
  scale_fill_manual(values = c("#1e8449", "#d79921", "#cc241d")) +  
  labs(  
    title = "MPG by cylinder count",  
    x = "Cylinders", y = "MPG"  
  ) +  
  theme_minimal(base_size = 11) +  
  theme(legend.position = "none")
```



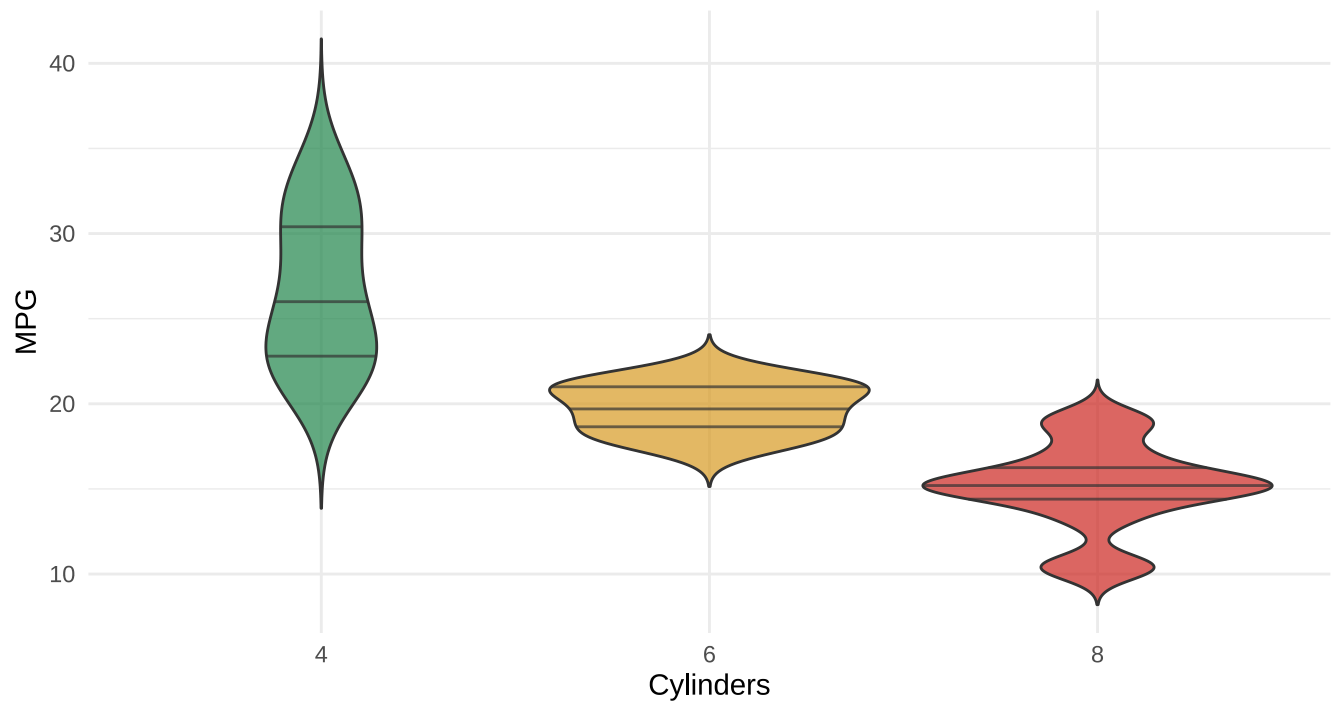
#### 4.3.5 Violin Plots (`geom_violin`)

**Use for:** full distribution shape + summary statistics – reveals multi-modality.

```
ggplot(mtcars, aes(factor(cyl), mpg, fill = factor(cyl))) +  
  geom_violin(  
    trim = FALSE,  
    draw_quantiles = c(0.25, 0.5, 0.75), alpha = 0.7  
  ) +  
  scale_fill_manual(values = c("#1e8449", "#d79921", "#cc241d")) +  
  labs(  
    title = "Violin plot of MPG by cylinders",  
    subtitle = "Quartile lines (25%, 50%, 75%) superimposed",  
    x = "Cylinders", y = "MPG"  
  ) +  
  theme_minimal(base_size = 11) +  
  theme(legend.position = "none")
```

## Violin plot of MPG by cylinders

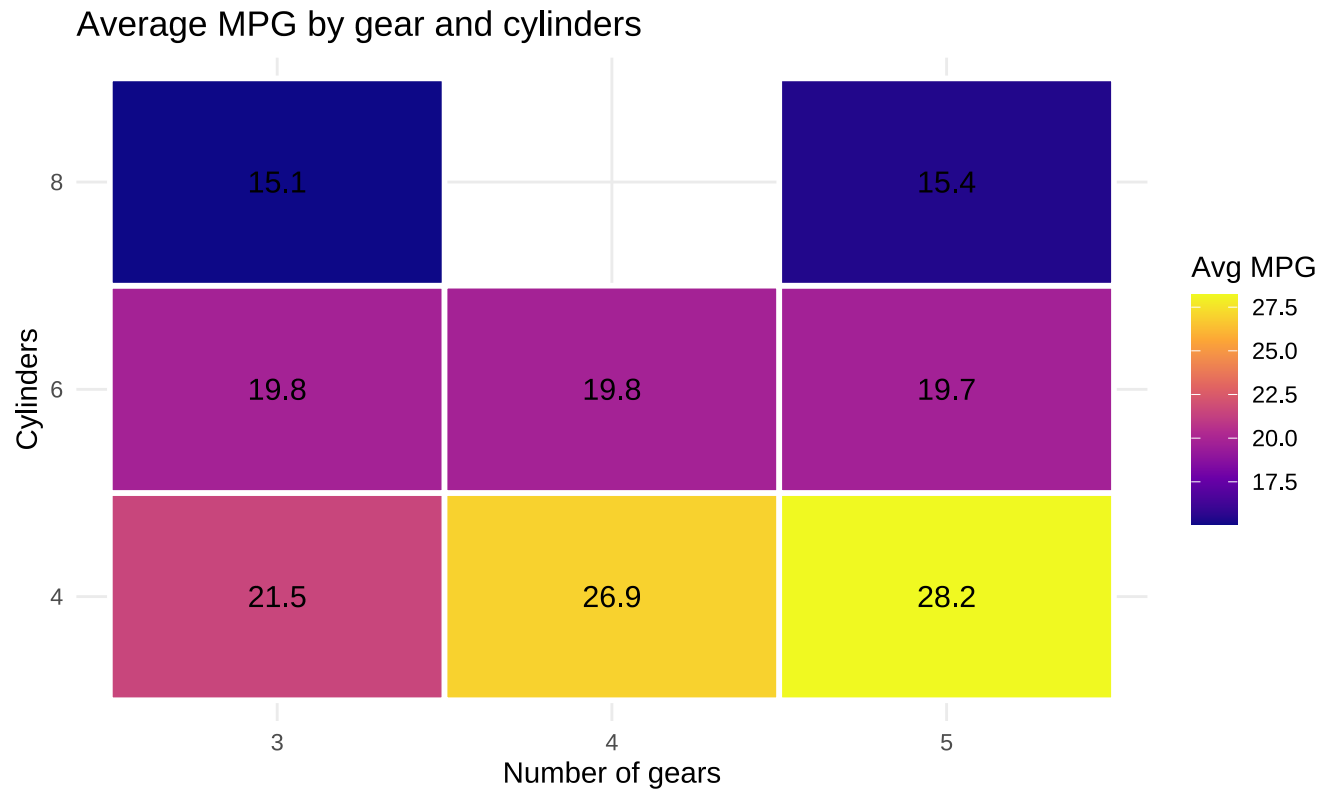
Quartile lines (25%, 50%, 75%) superimposed



### 4.3.6 Heatmaps (geom\_tile)

Use for: matrix of numeric values encoded by colour.

```
heat_data <- mtcars %>%  
  group_by(gear, cyl) %>%  
  summarise(avg_mpg = mean(mpg), .groups = "drop")  
  
ggplot(heat_data, aes(factor(gear), factor(cyl), fill = avg_mpg)) +  
  geom_tile(color = "white", linewidth = 1) +  
  geom_text(aes(label = round(avg_mpg, 1)), size = 4) +  
  scale_fill_viridis_c(option = "plasma", name = "Avg MPG") +  
  labs(  
    title = "Average MPG by gear and cylinders",  
    x = "Number of gears", y = "Cylinders"  
  ) +  
  theme_minimal(base_size = 11)
```



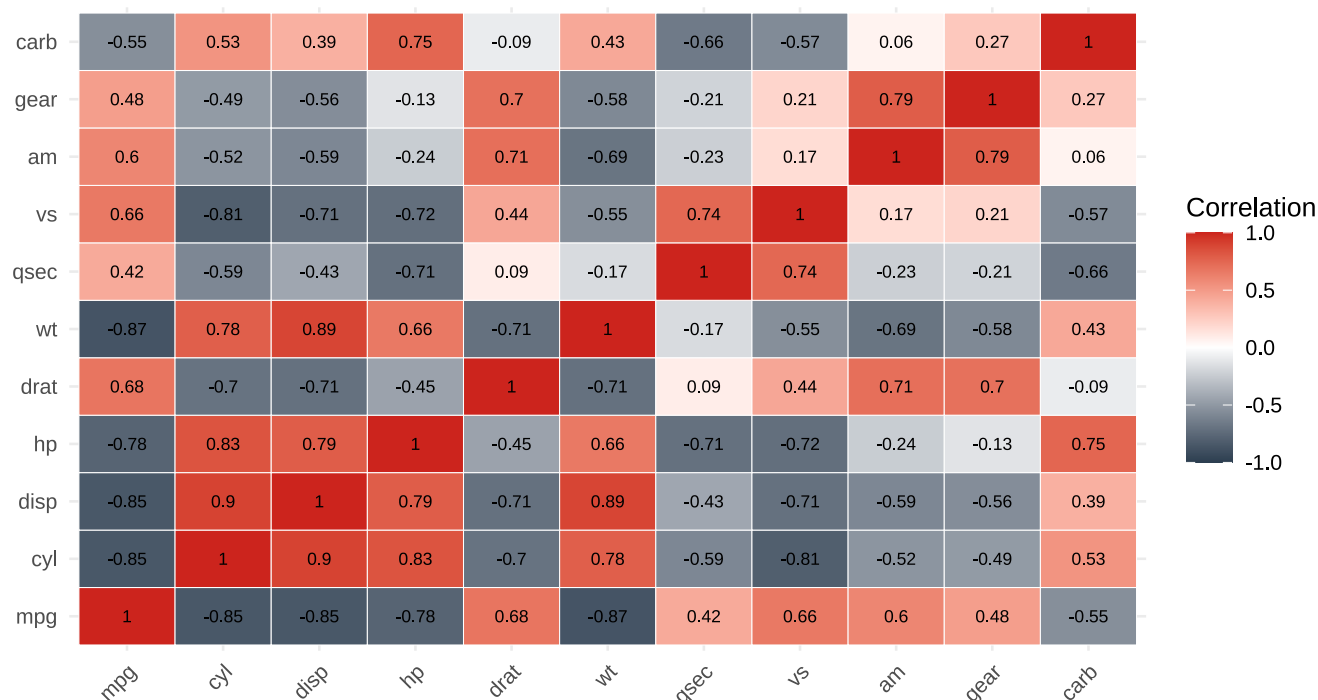
#### 4.3.7 Correlation Matrices

Use for: pairwise correlations between many numeric variables.

```
corr_matrix <- round(cor(mtcars), 2)
corr_long <- as.data.frame(as.table(corr_matrix))
names(corr_long) <- c("Var1", "Var2", "value")

ggplot(corr_long, aes(Var1, Var2, fill = value)) +
  geom_tile(color = "white") +
  geom_text(aes(label = value), size = 2.3) +
  scale_fill_gradient2(
    low = "#2c3e50", mid = "white", high = "#cc241d",
    midpoint = 0, limits = c(-1, 1),
    name = "Correlation"
  ) +
  labs(
    title = "Correlation matrix of mtcars",
    subtitle = "Values range from -1 (negative) to +1 (positive)"
  ) +
  theme_minimal(base_size = 10) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    axis.title = element_blank()
  )
)
```

Correlation matrix of mtcars  
Values range from -1 (negative) to +1 (positive)



#### 4.3.8 Plot Selection Guide

| Goal                                        | Plot                 |
|---------------------------------------------|----------------------|
| Relationship between 2 continuous variables | Scatter              |
| Distribution of 1 continuous variable       | Histogram or Density |
| Compare distributions across groups         | Boxplot or Violin    |
| 2-D aggregated values                       | Heatmap              |
| Pairwise correlations                       | Correlation matrix   |
| Time trend                                  | Line plot            |
| Part-to-whole                               | Stacked bar or Pie   |

## 5 Part IV – Correlation & Feature Relationships

### 5.1 Pearson Correlation

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

- Range:  $-1 \leq r \leq +1$ .
- $r = +1$ : perfect positive;  $r = -1$ : perfect negative;  $r = 0$ : no *linear* relationship.

| $ r $   | Interpretation |  |
|---------|----------------|--|
| 0.0–0.2 | Very weak      |  |
| 0.2–0.4 | Weak           |  |
| 0.4–0.6 | Moderate       |  |
| 0.6–0.8 | Strong         |  |
| 0.8–1.0 | Very strong    |  |

**Correlation causation.** Just because two variables are correlated does not mean one causes the other. There may be a confounding variable, or the relationship may be coincidental.

## 5.2 Spearman Correlation

**Spearman's rank correlation**  $\rho$  measures *monotonic* relationships (not necessarily linear).

- Based on ranks, not raw values.
- Robust to outliers.
- Use when the relationship is monotonic but non-linear.

## 5.3 Multicollinearity

**Multicollinearity** occurs when predictors are highly correlated with each other.

- **Problems:** unstable coefficients, inflated standard errors, difficult interpretation.
- **Detection:** correlation matrix ( $|r| > 0.8$  between predictors); Variance Inflation Factor ( $VIF > 5$  or  $10$ ).
- **Solution:** remove one of the correlated predictors; use Ridge regression; use PCA.

# 6 Part V – Regression Analysis

## 6.1 Simple Linear Regression (SLR)

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- $Y$  – dependent variable
- $X$  – independent variable
- $\beta_0$  – intercept
- $\beta_1$  – slope (change in  $Y$  per unit change in  $X$ )
- $\varepsilon$  – error term

**Least squares** finds the line minimising  $SSE = \sum (y_i - \hat{y}_i)^2$ .

### 6.1.1 Assumptions (LINE)

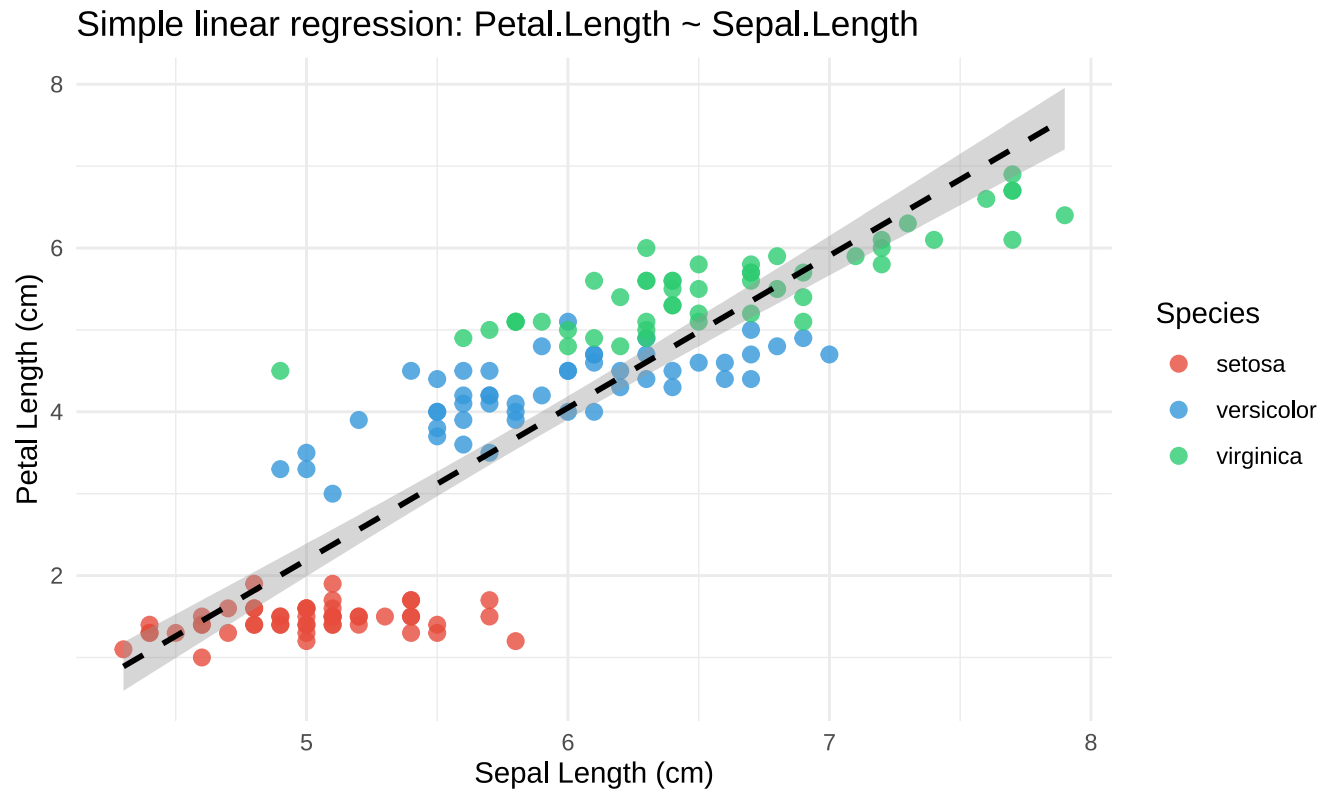
1. **Linearity**
2. **Independence**
3. **Normality of errors**
4. **Equal variance** (homoscedasticity)

### 6.1.2 Worked Example – Iris

```
model_slr <- lm(Petal.Length ~ Sepal.Length, data = iris)
summary(model_slr)
```

```
##
## Call:
## lm(formula = Petal.Length ~ Sepal.Length, data = iris)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.47747 -0.59072 -0.00668  0.60484  2.49512
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -7.10144     0.50666  -14.02  <2e-16 ***
## Sepal.Length   1.85843     0.08586   21.65  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8678 on 148 degrees of freedom
## Multiple R-squared:  0.76, Adjusted R-squared:  0.7583
## F-statistic: 468.6 on 1 and 148 DF, p-value: < 2.2e-16
```

```
ggplot(iris, aes(Sepal.Length, Petal.Length)) +
  geom_point(aes(color = Species), size = 2.5, alpha = 0.8) +
  geom_smooth(
    method = "lm", se = TRUE,
    color = "black", linetype = "dashed"
  ) +
  scale_color_manual(values = c("#E74C3C", "#3498DB", "#2ECC71")) +
  labs(
    title = "Simple linear regression: Petal.Length ~ Sepal.Length",
    x = "Sepal Length (cm)", y = "Petal Length (cm)"
  ) +
  theme_minimal(base_size = 11)
```



**How to read the summary:** The slope for Sepal.Length is roughly **+1.86** – every extra 1 cm of sepal length is associated with about 1.86 cm more petal length. The R-squared tells us what fraction of variation in petal length is explained by sepal length alone.

## 6.2 Multiple Linear Regression (MLR)

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \varepsilon$$

Each  $\beta_j$  is the change in  $Y$  for a one-unit change in  $X_j$ , **holding all other predictors constant**.

```
model_mlr <- lm(Petal.Length ~ Sepal.Length + Sepal.Width + Petal.Width,
  data = iris
)
summary(model_mlr)
```

```
##
## Call:
## lm(formula = Petal.Length ~ Sepal.Length + Sepal.Width + Petal.Width,
##     data = iris)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.99333 -0.17656 -0.01004  0.18558  1.06909
##
```

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.26271    0.29741  -0.883    0.379
## Sepal.Length  0.72914    0.05832  12.502   <2e-16 ***
## Sepal.Width  -0.64601    0.06850  -9.431   <2e-16 ***
## Petal.Width   1.44679    0.06761  21.399   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.319 on 146 degrees of freedom
## Multiple R-squared:  0.968, Adjusted R-squared:  0.9674
## F-statistic: 1473 on 3 and 146 DF, p-value: < 2.2e-16
```

### 6.2.1 Adjusted R-Squared

$$R_{adj}^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}$$

Always use **Adjusted**  $R^2$  for model comparison – adding useless predictors increases  $R^2$  but decreases Adjusted  $R^2$ .

---

## 6.3 Polynomial Regression

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \dots + \varepsilon$$

**Use when:** the relationship is curved. **Caution:** high-degree polynomials overfit.

```
model_poly <- lm(Petal.Length ~ Sepal.Length + I(Sepal.Length^2), data = iris)
summary(model_poly)$r.squared
```

```
## [1] 0.7773925
```

---

## 6.4 Logistic Regression

$$P(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p)}}$$

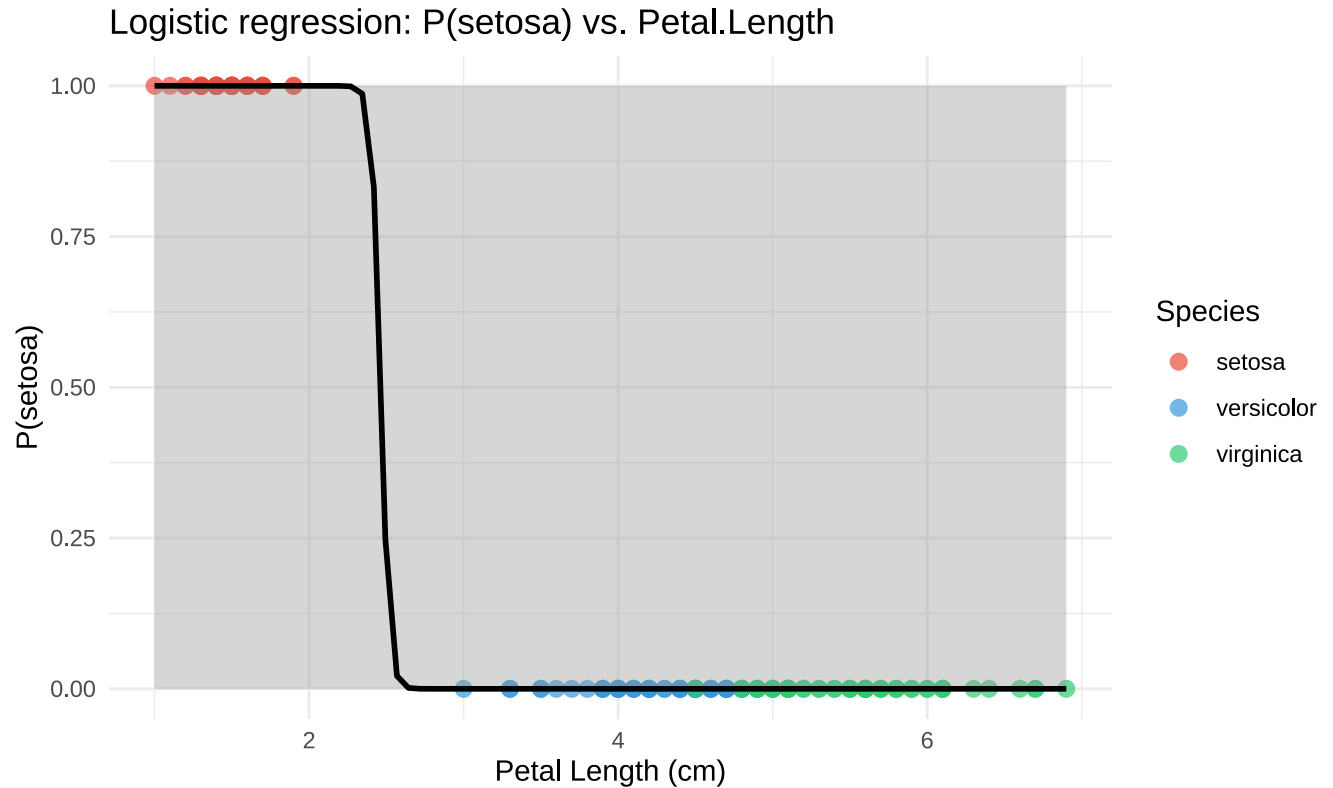
**Use when:** response variable is **binary**.

```
iris$is_setosa <- ifelse(iris$Species == "setosa", 1, 0)
model_logit <- glm(is_setosa ~ Petal.Length, data = iris, family = binomial)
summary(model_logit)
```

```
##
## Call:
## glm(formula = is_setosa ~ Petal.Length, family = binomial, data = iris)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)    91.67   47334.35   0.002   0.998
## Petal.Length  -37.22   18357.58  -0.002   0.998
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1.9095e+02  on 149  degrees of freedom
## Residual deviance: 7.3324e-09  on 148  degrees of freedom
## AIC: 4
##
## Number of Fisher Scoring iterations: 25
```

```
iris$prob_setosa <- predict(model_logit, type = "response")

ggplot(iris, aes(Petal.Length, prob_setosa)) +
  geom_point(aes(color = Species), size = 2.5, alpha = 0.7) +
  geom_smooth(
    method = "glm",
    method.args = list(family = binomial),
    color = "black", se = TRUE
  ) +
  scale_color_manual(values = c("#E74C3C", "#3498DB", "#2ECC71")) +
  labs(
    title = "Logistic regression: P(setosa) vs. Petal.Length",
    x = "Petal Length (cm)", y = "P(setosa)"
  ) +
  theme_minimal(base_size = 11)
```



## 6.5 Linear Discriminant Analysis (LDA)

LDA finds linear combinations of features that best separate classes.

### 6.5.1 Assumptions

1. **Normality** – each class is multivariate normal.
2. **Equal covariance matrices** across classes.
3. **Independence** of observations.

### 6.5.2 The Mathematics

LDA seeks  $w$  maximising  $J(w) = \frac{w^T S_B w}{w^T S_W w}$ , where  $S_B$  is between-class and  $S_W$  is within-class scatter.

**Intuition:** LDA finds the “best viewing angle” to separate the species. Imagine rotating a 3D scatter plot until the three species are most clearly separated – that is what LDA does mathematically.

```
model_lda <- lda(Species ~ Sepal.Length + Sepal.Width + Petal.Length + Petal.Width,
  data = iris
)

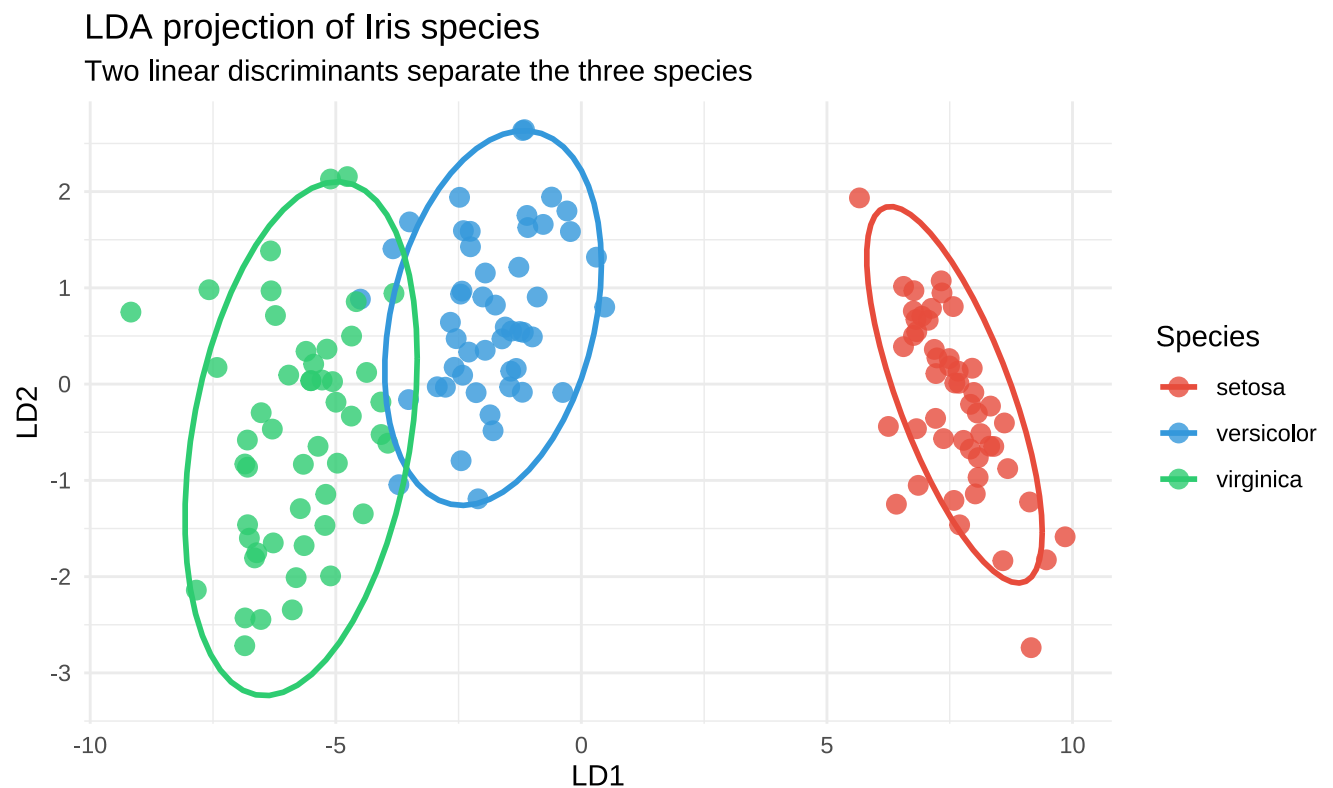
lda_scores <- predict(model_lda)$x
lda_df <- data.frame(
```

```

LD1      = lda_scores[, 1],
LD2      = lda_scores[, 2],
Species  = iris$Species
)

ggplot(lda_df, aes(LD1, LD2, color = Species)) +
  geom_point(size = 3, alpha = 0.8) +
  stat_ellipse(level = 0.95, linewidth = 1) +
  scale_color_manual(values = c("#E74C3C", "#3498DB", "#2ECC71")) +
  labs(
    title = "LDA projection of Iris species",
    subtitle = "Two linear discriminants separate the three species",
    x = "LD1", y = "LD2"
  ) +
  theme_minimal(base_size = 11)

```



```

lda_pred <- predict(model_lda)$class
conf_mat <- table(Predicted = lda_pred, Actual = iris$Species)
acc <- sum(diag(conf_mat)) / sum(conf_mat)
cat("LDA accuracy:", round(acc * 100, 2), "%\n")

```

```
## LDA accuracy: 98 %
```

```
conf_mat
```

```
##           Actual
## Predicted  setosa versicolor virginica
##   setosa      50         0         0
##   versicolor  0         48         1
##   virginica   0         2         49
```

---

## 6.6 Ridge, Lasso and Quantile Regression

### 6.6.1 Ridge (L2)

$$\text{Minimize: } \sum (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \beta_j^2$$

- Use when predictors are highly correlated.
- Shrinks coefficients toward zero but **never to exactly zero**.

### 6.6.2 Lasso (L1)

$$\text{Minimize: } \sum (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

- Performs **feature selection** – some coefficients become exactly zero.

### 6.6.3 Quantile Regression

Estimates conditional quantiles (median, 10th, 90th percentile) rather than the mean. **Robust to outliers**; ideal for extreme-value studies.

---

## 6.7 Model Comparison and Selection

| Method          | Type           | Response    | Key feature          | Best use              |
|-----------------|----------------|-------------|----------------------|-----------------------|
| Simple Linear   | Regression     | Continuous  | One predictor        | Basic relationships   |
| Multiple Linear | Regression     | Continuous  | Multiple predictors  | Complex relationships |
| Polynomial      | Regression     | Continuous  | Curved relationships | Non-linear patterns   |
| Logistic        | Classification | Binary      | Probability          | Binary outcomes       |
| LDA             | Classification | Categorical | Dimension reduction  | Multi-class problems  |
| Ridge           | Regression     | Continuous  | L2 regularization    | Correlated predictors |

| Method   | Type       | Response   | Key feature        | Best use          |
|----------|------------|------------|--------------------|-------------------|
| Lasso    | Regression | Continuous | L1 regularization  | Feature selection |
| Quantile | Regression | Continuous | Robust to outliers | Non-normal errors |

### 6.7.1 Decision Tree

#### 1. Response continuous?

- Yes  $\rightarrow$  2
- No (categorical)  $\rightarrow$  Logistic or LDA.

#### 2. One predictor?

- Yes  $\rightarrow$  Simple Linear.
- No  $\rightarrow$  3.

#### 3. Linear relationship?

- Yes  $\rightarrow$  Multiple Linear.
- No  $\rightarrow$  Polynomial or transform.

#### 4. Predictors highly correlated?

- Yes  $\rightarrow$  Ridge.
- Need feature selection  $\rightarrow$  Lasso.

#### 5. Outliers / non-normal errors?

- Yes  $\rightarrow$  Quantile or robust methods.

## 7 Part VI – Case Studies

### 7.1 Case Study 1 – Rainfall Variability (Synthetic weatherAUS)

**Why synthetic data?** The original dataset (`weatherAUS.csv`) is not shipped with this handout. We generate a realistic **synthetic** dataset that mimics the statistical properties of the Sydney observations so that every code chunk knits standalone.

#### 7.1.1 Load and Clean

```
set.seed(42)
n_days <- 800
dates <- seq(as.Date("2011-01-01"), by = "day", length.out = n_days)

# Simulate realistic Sydney weather
sydney <- data.frame(
  Date      = dates,
  MinTemp   = rnorm(n_days, 13, 5),
  MaxTemp   = rnorm(n_days, 23, 6),
  Humidity9am = pmin(pmax(rnorm(n_days, 70, 15), 10), 100),
  Humidity3pm = pmin(pmax(rnorm(n_days, 55, 18), 10), 100),
```

```

WindSpeed9am = abs(rnorm(n_days, 15, 8)),
Pressure9am = rnorm(n_days, 1015, 7)
)

# Synthetic rainfall: skewed, correlated with humidity and pressure
sydney$Rainfall <- with(
  sydney,
  pmax(0, 0.03 * Humidity3pm - 0.01 * Pressure9am + rnorm(n_days, 2, 3))
)

sydney$AvgTemp <- (sydney$MinTemp + sydney$MaxTemp) / 2
sydney$logRainfall <- log(sydney$Rainfall + 1)

cat("Complete days:", nrow(sydney), "\n")

```

```
## Complete days: 800
```

```
head(sydney, 3)
```

```
##      Date MinTemp MaxTemp Humidity9am Humidity3pm WindSpeed9am Pressure9am
## 1 2011-01-01 19.85479 27.13285    70.43975    78.89508     1.516208    1014.007
## 2 2011-01-02 10.17651 27.35050    99.01472    44.18496    20.417994    1009.303
## 3 2011-01-03 14.81564 24.30428    82.56266    56.01712    21.691064    1012.721
##  Rainfall AvgTemp logRainfall
## 1         0 23.49382         0
## 2         0 18.76350         0
## 3         0 19.55996         0
```

### 7.1.2 Distributions

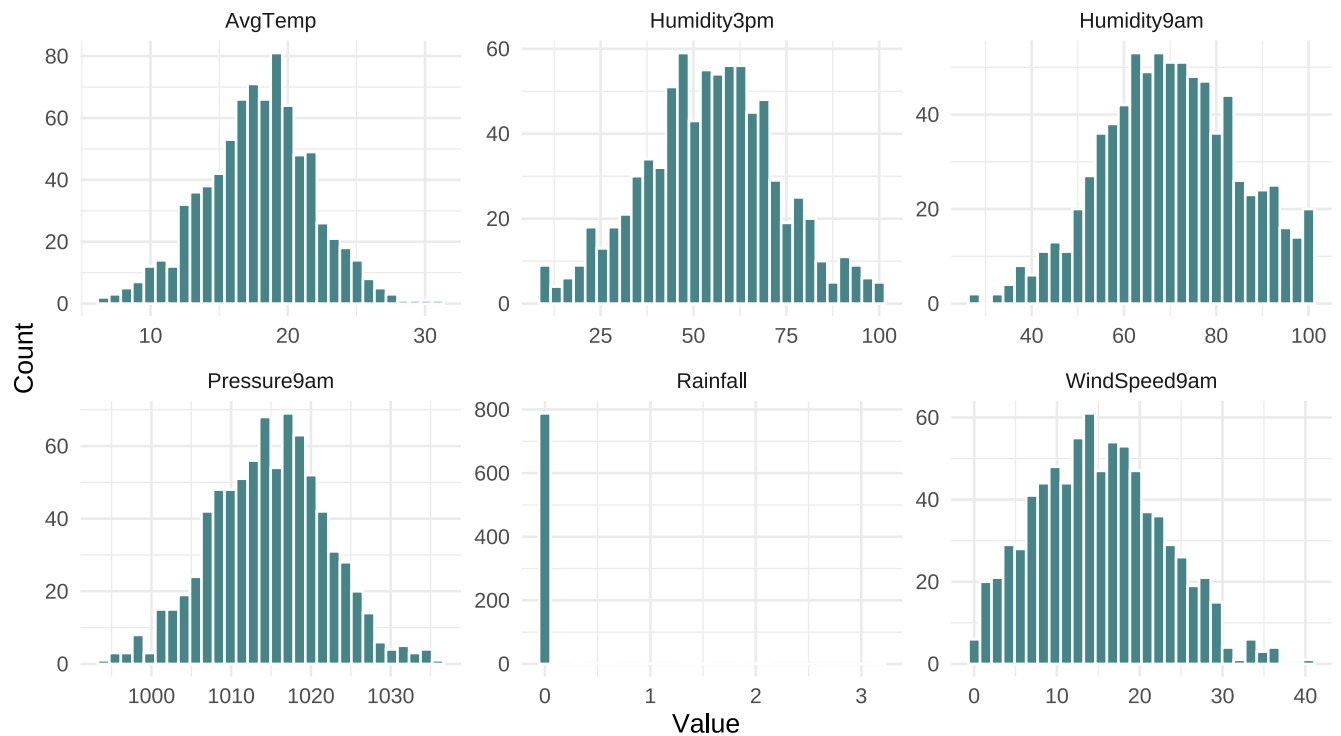
```

vars <- c(
  "Rainfall", "AvgTemp", "Humidity9am", "Humidity3pm",
  "WindSpeed9am", "Pressure9am"
)
syd_long <- pivot_longer(sydney, all_of(vars),
  names_to = "Variable", values_to = "Value"
)

ggplot(syd_long, aes(Value)) +
  geom_histogram(bins = 30, fill = "#458588", color = "white") +
  facet_wrap(~Variable, scales = "free") +
  labs(
    title = "Distributions of key weather variables (synthetic)",
    x = "Value", y = "Count"
  ) +
  theme_minimal(base_size = 10)

```

## Distributions of key weather variables (synthetic)



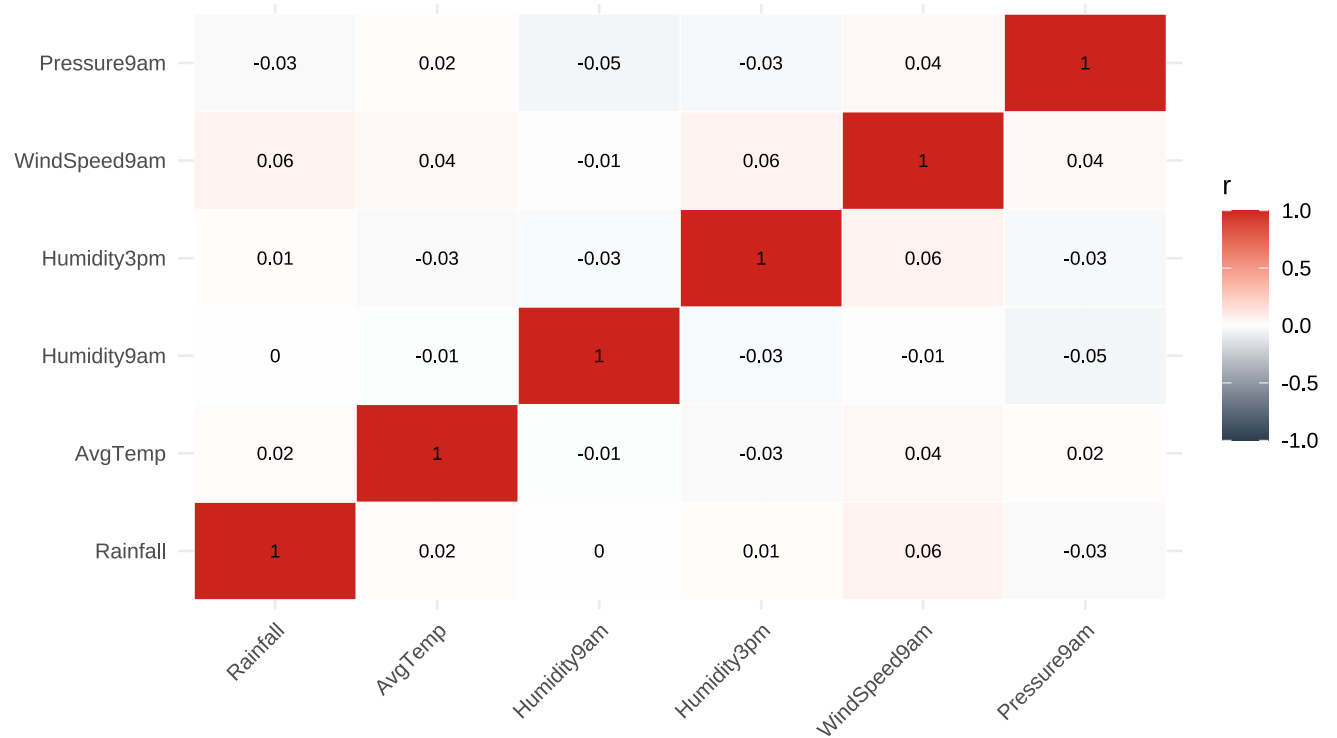
**How to read this figure:** Rainfall is heavily **right-skewed** – most days have little rain, but a few have downpours. Temperature is roughly bell-shaped. This skew justifies the **log-transform** for regression.

### 7.1.3 Correlation Heatmap

```
cor_matrix <- cor(sydney[, vars])
cor_long <- as.data.frame(as.table(cor_matrix))
names(cor_long) <- c("Var1", "Var2", "value")

ggplot(cor_long, aes(Var1, Var2, fill = value)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(value, 2)), size = 2.5) +
  scale_fill_gradient2(
    low = "#2c3e50", mid = "white",
    high = "#cc241d",
    midpoint = 0, limits = c(-1, 1),
    name = "r"
  ) +
  labs(title = "Correlation matrix -- Sydney weather (synthetic)") +
  theme_minimal(base_size = 10) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    axis.title = element_blank()
  )
)
```

Correlation matrix -- Sydney weather (synthetic)



#### 7.1.4 Multiple Regression on log(Rainfall)

```
model_rain <- lm(logRainfall ~ AvgTemp + Humidity3pm + Pressure9am,
  data = sydney
)
summary(model_rain)
```

```
##
## Call:
## lm(formula = logRainfall ~ AvgTemp + Humidity3pm + Pressure9am,
##     data = sydney)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.02237 -0.01259 -0.01037 -0.00825  1.41665
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.499e-01  4.923e-01   0.914   0.361
## AvgTemp      2.452e-04  8.854e-04   0.277   0.782
## Humidity3pm   6.066e-05  1.918e-04   0.316   0.752
## Pressure9am  -4.405e-04  4.846e-04  -0.909   0.364
##
## Residual standard error: 0.09849 on 796 degrees of freedom
## Multiple R-squared:  0.001267,    Adjusted R-squared:  -0.002497
```

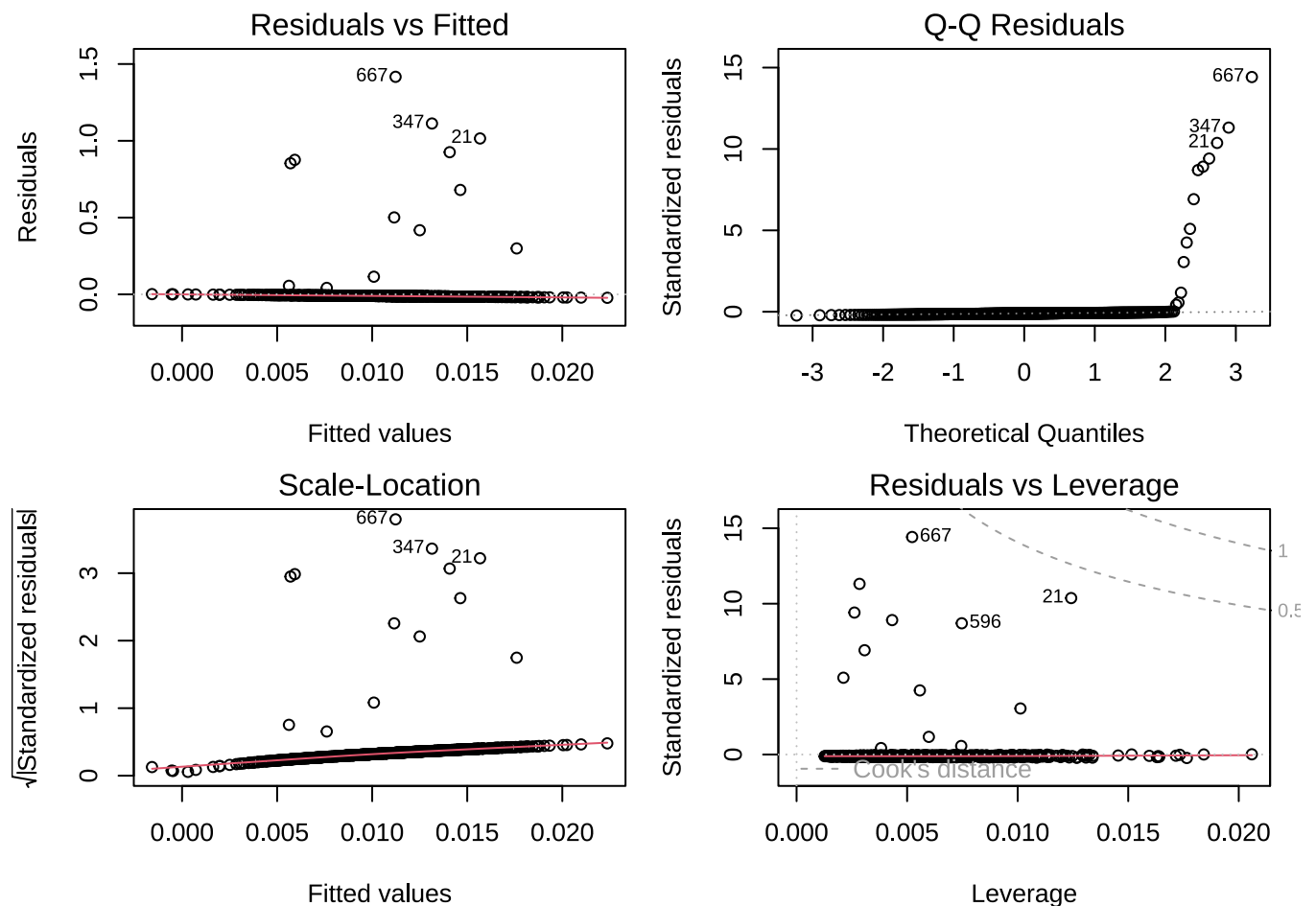
## F-statistic: 0.3366 on 3 and 796 DF, p-value: 0.7989

How to read the summary: The coefficients tell us:

- **Humidity3pm** – strongly *positive*: humid afternoons precede rain.
- **Pressure9am** – *negative*: low pressure = storm systems.
- **AvgTemp** – *negative*: warmer days tend to be drier.
- **Adjusted R-squared** tells us how much of the log-rainfall variability these three predictors jointly explain.

### 7.1.5 Diagnostics

```
par(mfrow = c(2, 2), mar = c(4, 4, 2, 1))
plot(model_rain, cex = 0.9)
```



```
par(mfrow = c(1, 1))
```

How to read diagnostics:

- **Residuals vs Fitted** – random scatter around 0 is good; a curve means non-linearity.
  - **Normal Q-Q** – points along the diagonal means normal errors.
  - **Scale-Location** – flat line means constant variance.
  - **Residuals vs Leverage** – points outside Cook's distance contours are influential.
- 

## 7.2 Case Study 2 – Retail Sales EDA (Synthetic Superstore)

As with Case Study 1, the original `Superstore.xlsx` is not shipped. We generate a **synthetic** orders dataset that preserves the qualitative patterns (yearly growth, category mix, discount effects) so that all code chunks are reproducible.

### 7.2.1 Load and Prepare

```
set.seed(7)
n <- 3000
regions <- c("West", "East", "Central", "South")
cats <- c("Furniture", "Office Supplies", "Technology")
subs <- c(
  "Chairs", "Tables", "Bookcases", "Furnishings",
  "Phones", "Machines", "Copiers", "Accessories",
  "Binders", "Storage", "Art", "Labels"
)

superstore <- data.frame(
  OrderDate = sample(seq(as.Date("2011-01-01"),
    as.Date("2014-12-31"),
    by = "day"), n, replace = TRUE),
  Region = sample(regions, n,
    replace = TRUE,
    prob = c(0.3, 0.3, 0.22, 0.18)
  ),
  Category = sample(cats, n, replace = TRUE),
  SubCategory = sample(subs, n, replace = TRUE),
  Quantity = sample(1:10, n, replace = TRUE),
  Discount = sample(c(0, 0.1, 0.2, 0.3, 0.4), n,
    replace = TRUE, prob = c(.5, .2, .15, .1, .05)
  ),
  Sales = round(runif(n, 20, 900), 2)
)

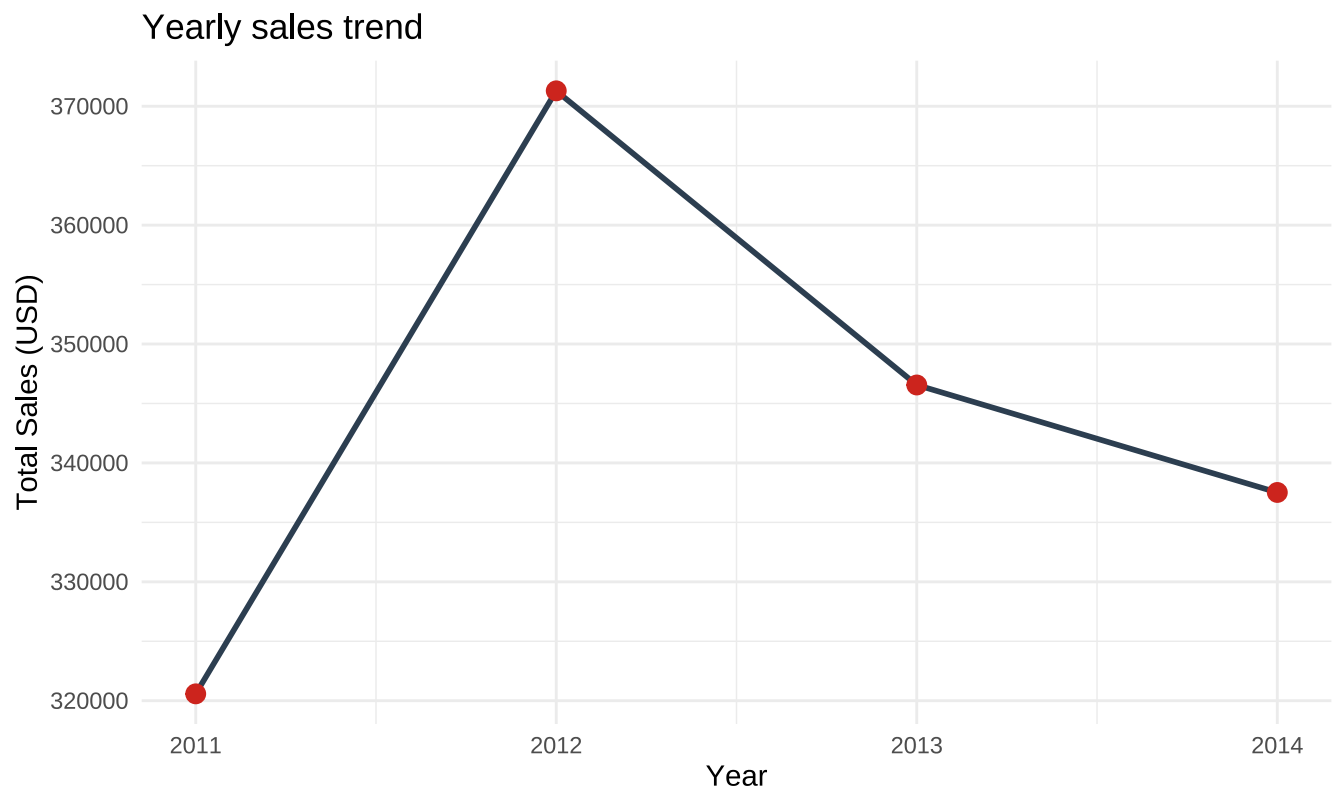
# Realistic profit: negative on heavy discounts, category-dependent
superstore$Profit <- round(
  superstore$Sales * (0.25 - 0.6 * superstore$Discount) +
  rnorm(n, 0, 15), 2
)

superstore$Year <- as.integer(format(superstore$OrderDate, "%Y"))
superstore$Month <- as.integer(format(superstore$OrderDate, "%m"))
superstore$ProfitMargin <- superstore$Profit / superstore$Sales * 100
```

### 7.2.2 Yearly Sales

```
yearly <- superstore %>%
  group_by(Year) %>%
  summarise(
    total_sales = sum(Sales), orders = n(),
    .groups = "drop"
  )

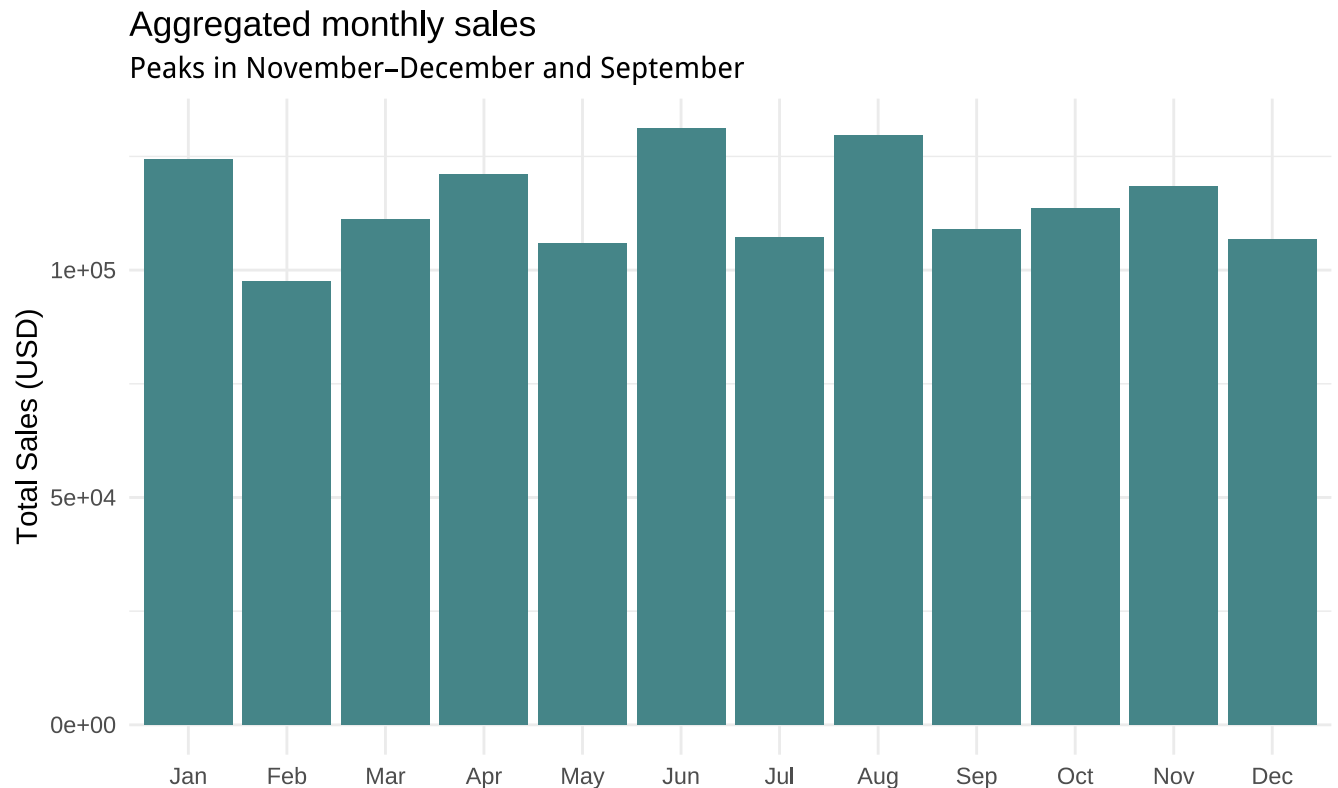
ggplot(yearly, aes(Year, total_sales)) +
  geom_line(color = "#2c3e50", linewidth = 1) +
  geom_point(size = 3, color = "#cc241d") +
  labs(
    title = "Yearly sales trend",
    x = "Year", y = "Total Sales (USD)"
  ) +
  theme_minimal(base_size = 11)
```



### 7.2.3 Monthly Seasonality

```
monthly <- superstore %>%
  group_by(Month) %>%
  summarise(total_sales = sum(Sales), .groups = "drop")
```

```
ggplot(monthly, aes(factor(Month, labels = month.abb), total_sales)) +
  geom_col(fill = "#458588") +
  labs(
    title = "Aggregated monthly sales",
    subtitle = "Peaks in November-December and September",
    x = NULL, y = "Total Sales (USD)"
  ) +
  theme_minimal(base_size = 11)
```



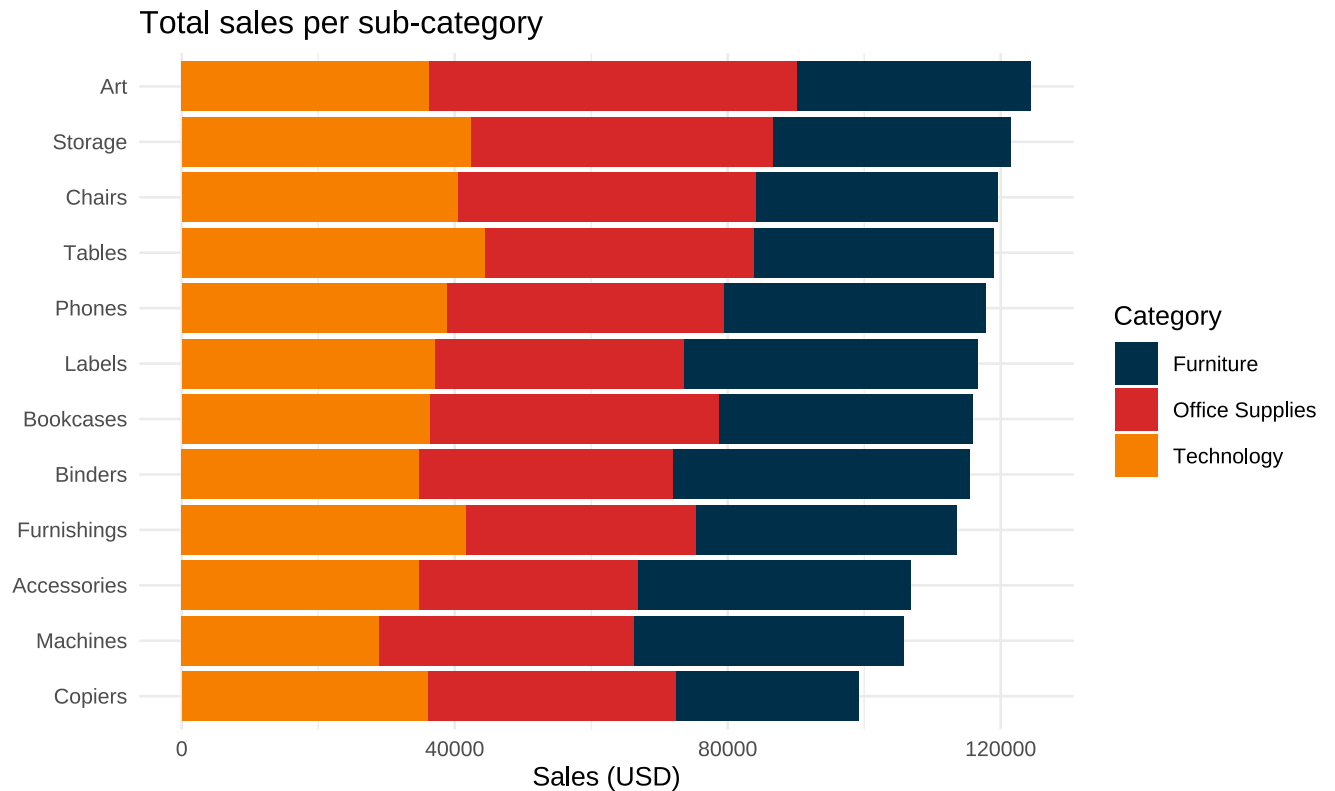
**How to read:** Sales peak in the holiday season (Nov–Dec) and around the start of the school year (Sept). Retailers plan stock and staffing around these peaks.

## 7.2.4 Sales by Sub-Category

```
subcat_sales <- superstore %>%
  group_by(Category, SubCategory) %>%
  summarise(Sales = sum(Sales), .groups = "drop") %>%
  arrange(desc(Sales))

ggplot(
  subcat_sales,
  aes(reorder(SubCategory, Sales), Sales, fill = Category)
) +
```

```
geom_col() +
coord_flip() +
scale_fill_manual(values = c("#003049", "#d62828", "#f77f00")) +
labs(
  title = "Total sales per sub-category",
  x = NULL, y = "Sales (USD)"
) +
theme_minimal(base_size = 10)
```



### 7.2.5 Profitability

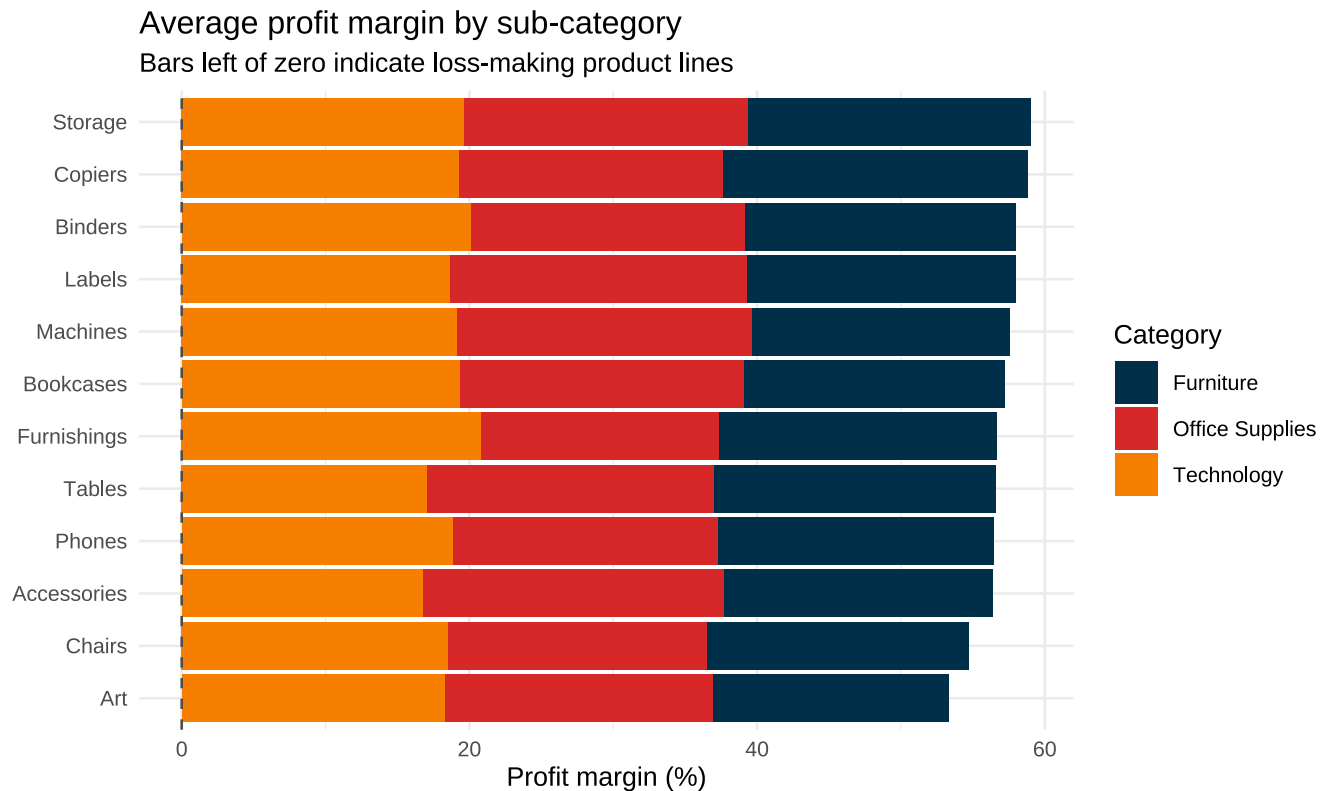
```
profit_summary <- superstore %>%
  group_by(Category, SubCategory) %>%
  summarise(Margin = mean(ProfitMargin, na.rm = TRUE), .groups = "drop")

ggplot(
  profit_summary,
  aes(Margin, reorder(SubCategory, Margin), fill = Category)
) +
geom_col() +
scale_fill_manual(values = c("#003049", "#d62828", "#f77f00")) +
geom_vline(xintercept = 0, linetype = "dashed", color = "grey30") +
labs(
  title = "Average profit margin by sub-category",
```

```

    subtitle = "Bars left of zero indicate loss-making product lines",
    x = "Profit margin (%)", y = NULL
  ) +
  theme_minimal(base_size = 10)

```



**Key insight:** Some product lines (typically those with heavy discounts) sit **left of zero** – they are loss-making. Re-evaluating the discount policy on these lines is usually the fastest route to profit improvement.

## 7.2.6 Discount Impact

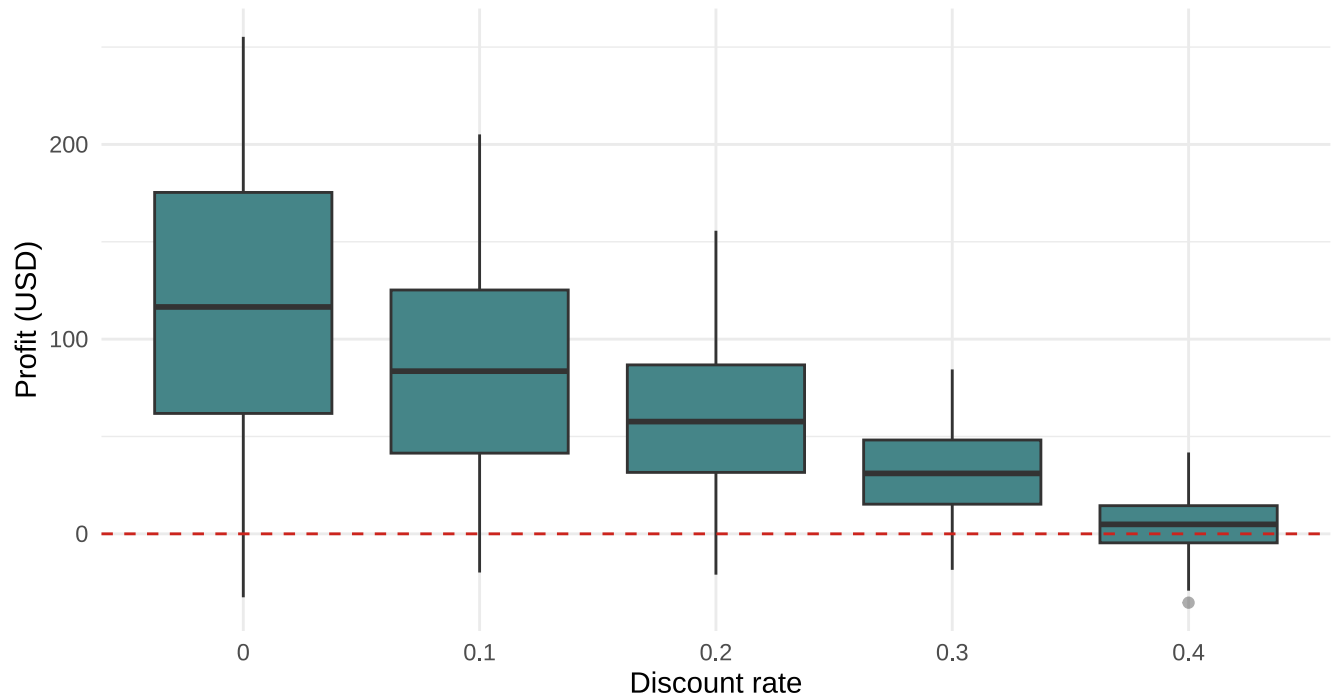
```

ggplot(superstore, aes(factor(Discount), Profit)) +
  geom_boxplot(fill = "#458588", outlier.alpha = 0.4) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "#cc241d") +
  labs(
    title = "Profit vs. discount level",
    subtitle = "Higher discounts push profit toward or below zero",
    x = "Discount rate", y = "Profit (USD)"
  ) +
  theme_minimal(base_size = 11)

```

## Profit vs. discount level

Higher discounts push profit toward or below zero



### 7.2.7 Key Recommendations

1. **Focus on high-performing products** – invest in the top-selling sub-categories.
2. **Optimise pricing and discounts** – reduce discounts on loss-making products.
3. **Regional strategies** – target South and Central for growth.
4. **Seasonal planning** – stock up for November, December, September.
5. **Product mix optimisation** – emphasise high-margin items.
6. **Invest in analytics** – real-time dashboards for monitoring.

## 8 Appendix A – Quick Reference Tables

### 8.1 A.1 Descriptive Statistics Formulas

| Measure            | Formula                                      | Robust? |
|--------------------|----------------------------------------------|---------|
| Mean               | $\bar{x} = \frac{1}{n} \sum x_i$             | No      |
| Median             | Middle value                                 | Yes     |
| Variance           | $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$ | No      |
| Standard Deviation | $s = \sqrt{s^2}$                             | No      |
| IQR                | $Q_3 - Q_1$                                  | Yes     |
| Range              | Max – Min                                    | No      |
| CV                 | $\frac{s}{\bar{x}} \times 100\%$             | No      |

## 8.2 A.2 Outlier Detection

| Method  | Rule                                                                      | Assumption |
|---------|---------------------------------------------------------------------------|------------|
| IQR     | $x < Q_1 - 1.5 \times \text{IQR}$ or<br>$x > Q_3 + 1.5 \times \text{IQR}$ | None       |
| Z-score | $\ z\  > 3$ , with<br>$z = (x - \bar{x})/s$                               | Normality  |

## 8.3 A.3 Regression Assumptions (LINE)

|                | Assumption           | Check | Fix                    |
|----------------|----------------------|-------|------------------------|
| Linearity      | Residuals vs. Fitted |       | Add polynomial terms   |
| Independence   | Residuals vs. order  |       | Time-series models     |
| Normality      | Q-Q plot             |       | Transform $Y$          |
| Equal variance | Scale-Location       |       | Weighted least squares |

## 8.4 A.4 Model Selection Guide

| Scenario                                 | Recommended method         |
|------------------------------------------|----------------------------|
| Continuous response, 1 predictor         | Simple Linear Regression   |
| Continuous response, multiple predictors | Multiple Linear Regression |
| Curved relationship                      | Polynomial Regression      |
| Binary response                          | Logistic Regression        |
| Multi-class classification               | LDA                        |
| Multicollinearity                        | Ridge Regression           |
| Feature selection                        | Lasso Regression           |
| Outliers / non-normal errors             | Quantile Regression        |

## 9 Appendix B – Mini Glossary

- **Feature** – an input column the model learns from (cement, rainfall intensity).
- **Label / Target** – what we predict (strength, Flood/No Flood).
- **Training set** – data the model learns from. **Test set** – data used for honest grading.
- **Imputation** – filling missing values (e.g., with the median).
- **Normalisation / Standardisation** – rescaling features to comparable ranges.
- **One-hot encoding** – one 1/0 column per category.
- **Pearson / Spearman** – linear vs. rank-based correlation.
- **Multicollinearity** – high correlation among predictors.
- **Ridge / Lasso** – L2 / L1 regularisation.
- **LDA** – Linear Discriminant Analysis (classification + reduction).
- **Overfitting / Underfitting** – model too complex / too simple.
- **Bias / Variance** – systematic error vs. instability.
- **STL** – Seasonal-Trend decomposition using Loess.

- **AAGR** – Average Annual Growth Rate.
  - **VIF** – Variance Inflation Factor.
- 

## 10 Appendix C – Exam Preparation Tips

1. **Understand concepts, don't memorize formulas.** Know *why* a method is used.
  2. **Practice interpreting output.** Given a regression summary, explain each coefficient.
  3. **Link plots to purposes.** Know when to use each chart type and what to look for.
  4. **Know the assumptions.** Every method has them; know how to check them.
  5. **Think about the workflow.** EDA → Cleaning → Visualisation → Modeling → Diagnostics → Interpretation.
  6. **Use examples** – iris, mtcars, rainfall, retail.
  7. **Distinguish correlation from causation.**
  8. **Parametric vs. non-parametric** – know when to use each.
  9. **R-squared vs. Adjusted R-squared** – why is the latter preferred?
  10. **LDA intuition** – the “best viewing angle” to separate classes.
- 

## 11 Further Reading

1. James, G., Witten, D., Hastie, T., and Tibshirani, R. (2021). *An Introduction to Statistical Learning* (2nd ed.). Springer.
2. Wickham, H., Cetinkaya-Rundel, M., and Grolemund, G. (2023). *R for Data Science* (2nd ed.). O'Reilly.
3. Hastie, T., Tibshirani, R., and Friedman, J. (2009). *The Elements of Statistical Learning* (2nd ed.). Springer.
4. Boehmke, B., and Greenwell, B. M. (2019). *Hands-On Machine Learning with R*. CRC Press.
5. Course notes 24CV201T, PDEU – Units 2 and 3.